

Assignments in **Biology** 3

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Peter Gribben



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Biology
3

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Peter Gribben



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Preface

This set of books has been designed to help teachers of biology in most aspects of this subject. Students will find the language, concepts and material much easier to use and assimilate than that in conventional textbooks. The content includes 'stand alone' material as well as sheets that require some existing knowledge and that test these concepts. An understanding of key phrases and concepts is encouraged.

A variety of styles has been used to engage students' attention. Whilst the books have been mainly aimed at senior students, teachers of able year nine and ten students will find much that they can use with these year groups. In order to extend the more motivated students, many topics include a research section. Outline answers are given for these research questions.

The material can be used as reinforcement, revision, homework, and as an introduction to new topics. Replacement teachers who may not be totally familiar with the subject will find these books invaluable.

Note: In most cases space has not been provided for students to answer the questions on the worksheets. Answers should be written in students' usual class work book or folder.

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What is life?

1

Characteristics of living things

Most of us can recognise the difference between living and non-living things. Living organisms are identified by a number of characteristics. They:

- move, or are in some way active (eg sessile (non-moving) organisms actively take up nutrients)
- have a recognisable size and shape (per species)
- respond to, and play an active role in, their external environment
- take up nutrients (organic compounds) from the environment
- reproduce ('life arises from life')
- have a reasonably predictable life span (eg die at a predictable 'old age' barring accident or early onset of disease)

'Life was never meant to be easy' is a very common statement. Living organisms must fight to survive. They do this using a number of means, including variation and adaptation.

- 1 Name five sessile organisms.
- 2 Discuss the strategies organisms use to survive. As well as mentioning 'variation' and 'adaptation', include another survival strategy.

Unicellular and multicellular organisms

Organisms, or their component parts, can be made up of a single cell or many cells. Single-celled organisms are called *unicellular* (*uni* one) while those composed of more than one cell are *multicellular* organisms (*multi* many). With so many cells involved, the cells of multicellular organisms have evolved to specialise, resulting in a division of labour. For example, sperm and eggs are specialised cells that are only involved in reproduction.

Cells with a common embryonic origin (in other words, similar cell types) can group together to share a common structure and function. This structure is called the *tissue* in an organism. Two or more types of tissue can work together to create a functional unit called an *organ*. Organs perform specific tasks, for example the heart acts as a pump to circulate blood. When two or more organs work together an *organ system* develops. The digestive system is an example of an organ system. Organ systems form an organism.

- 3 Identify four organs that are part of the digestive system.
- 4 Identify four organs that are part of the circulatory system.

Living or non-living?

There has been much debate as to whether certain things should be classified as living or non-living. Bacteria, fungi, viruses and prions have all been placed into both categories at various times. Bacteria and fungi are classified as living things but viruses and prions are generally regarded as non-living.

Viruses

Viruses have often been called 'the living dead', surrounding them with a science fiction-like mystery. Viruses are little more than units carrying genetic material (DNA or RNA) housed in a mechanism designed to insert it into a living cell. Without the cell, viruses are considered dormant, inactive. But given access to a living cell they attach to it and inject their genetic material into the cell. The cell is then taken over and used to replicate numerous copies of the virus. Eventually, so many are in the cell that it bursts open (lysis) and the flood of newly manufactured viruses repeat the procedure. Without access to a living cell, they remain inactive (for up to thousands of years, depending on the virus). However, once they have access to the cell's organelles, a necessary requirement to replication, they actively reproduce.

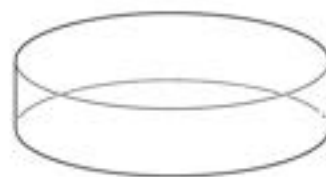
- 5 List five characteristics of a living thing.
- 6 Draw up a list that summarises the points for and against viruses being classified as living.
- 7 Give the full name for the acronym *DNA*.
- 8 Bacteria is the plural name for a single bacterium. What is the singular name for *fungi*?

Culturing bacteria

In classification, bacteria make up the kingdom known as *Monera*. There are thousands of different species of bacteria and they exist in many different types of environment. Sometimes scientists need to know if bacteria are present in a particular place and, if so, to identify them.

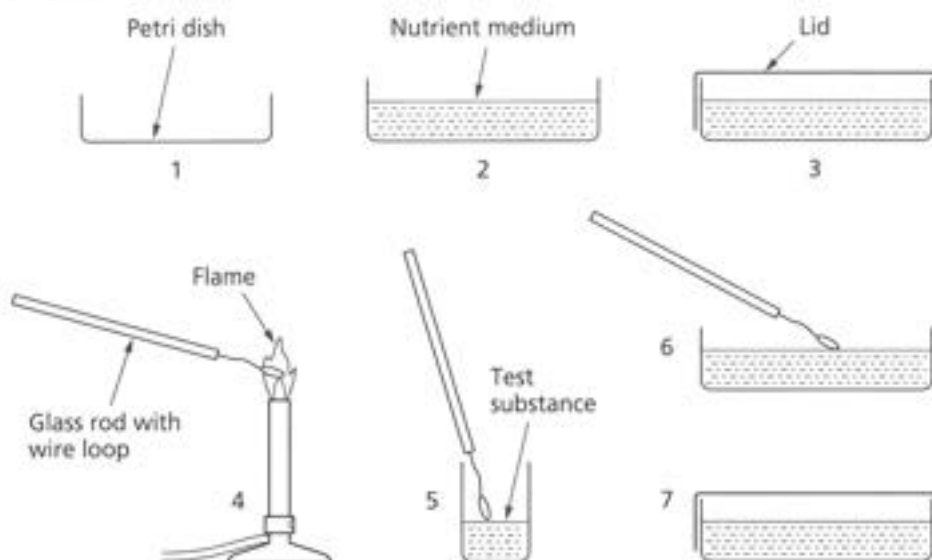
One technique that microbiologists use to grow bacteria (and other microorganisms) is *culturing*. A Petri dish (and lid) is a common piece of apparatus used in culturing bacteria. It is a shallow, circular dish with a cover and was invented in 1887 by R J Petri, a colleague of Robert Koch (1843–1910)*. Petri dishes are made from glass or clear plastic. Microorganisms are grown on a nutrient medium of agar gel. Agar-Agar is obtained from seaweed and contains a powerful gelatinising agent.

* Some sources state that Petri modified Koch's original design.



Petri dish

- 9 The diagrams show the sequence followed in culturing microorganisms. Describe how this process is carried out, referring to the diagrams.



- 10 Define *nutrient*.
 11 Define *gel*.
 12 Define *environment*.
 13 Name the types of organisms that make up the kingdom *Monera*.

Decide whether the following statements are true or false:

- 14 Bacteria replicate by cell division.
 15 Like viruses, bacteria are not considered to be true organisms.
 16 Bacteria are unicellular organisms.
 17 Bacteria are classified on the basis of their shape.

Research

- 18 *Legionnaires' disease* is an illness that was first reported in 1976. Describe the symptoms, name the causal agent, most common location of outbreaks of the disease, and outline the treatment for the disease.
 19 Outline some of Robert Koch's contributions to Medicine.

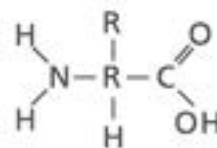
Prions: mysterious 'brain eating' proteins

When we think of disease-causing agents, most people think of bacteria, viruses, fungi or parasites. While these are often the cause of disease, the *prion* (pronounced *pree – on*) now presents a challenging alternative. The existence of disease-causing prions, which infect the brain, is a hotly debated issue. However, evidence suggests that they are real and that they do cause disease. The question is *how*?

What is a prion?

Prions are proteins that are made of about 250 amino acid groups. They are produced in the cell and are a part of our bodies. Not all prions behave in a 'natural', healthy way. Rogue prions have been identified. They have a recognisably deformed structure and are believed to cause the 'prion diseases'. The deformed prions, of which about 20 variations have been recognised to date, only differ from healthy ones in the way they are folded. The different shape appears to be linked to their ability to cause disease. (Imagine creating an animal shape using origami then, using exactly the same sheet of paper, opening it out and refolding it into another shape.)

Structure of an amino acid



(R = organic group)

- 1 Name the elements present in an amino acid.

Prion diseases

In 1986, England experienced its first 'Mad cow' epidemic. Mad cow, or Bovine spongiform encephalopathy – BSE, is an incurable disease that causes parts of the brain to be eaten away. This leads to coma and death within about 3 to 12 months of infection. The name for BSE (pronounced *bo – vine sponj – ee – form – en – sef – alo – path – ee*) is derived thus:

- *bovine*: refers to cows; *spongiform*: sponge-like; *encephalopathy*: brain degeneration.

The brain tissue of affected animals ends up hole-ridden and resembling a sponge. This perforation of the brain is a common feature shared by all diseases suspected of being caused by infective prions. In sheep, the disease commonly called 'scrapie' occurs. The human form is called nvCJD, which is a new variant of *Creutzfeldt-Jakob Disease*, or *CJD* (pronounced *croits – feld – yak – ob*). The lower-case 'nv' is added to the CJD to indicate that it is a new variant, believed to be caused by prions. Some people refer to it as 'human-form BSE'.

A connection has been made between cattle feed that contained the offal of sheep infected with scrapie and the cattle later developing BSE. Some humans who ate BSE-infected meat have developed nvCJD. Yet the actual infective prions have not yet been isolated and identified in the muscle of any infected beef to prove the connection. This leads researchers to question whether the prions, which clump in the brain, spread out within muscle tissue and are so limited in number that they are nearly impossible to identify. Experiments with mice found tiny pools of infective prions within muscle tissue but only under specific conditions, which may not mirror real life situations.

- 2 Describe the symptoms of a *coma*.
- 3 Give a definition for *offal*.
- 4 Identify the disease described here: 'A disease of the nervous system of sheep that has symptoms of twitching, severe thirst, loss of weight and paralysis.'
- 5 Give the full name for the acronym BSE.
- 6 Give the full name for the acronym CJD.
- 7 Anyone who ate beef in Britain in the 1980s is not allowed to donate blood in Australia. Give possible reasons for this.

Here are some abbreviations you will encounter when reading about prions:

- **PrPC**: Healthy prions found in cells (C: cell)
- **PrP**: Prion protein (*Pr*: Protein); indicates the abnormally folded prion linked to BSE
- **PrPSc**: Prions connected with the sheep disease 'scrapie' (Sc: Scrapie)



BSE-infected brains provide grim visual evidence that the infective agent was present. The deformed prions have been isolated from brain tissue and associated neurons. Prions have not yet been isolated in beef and, although samples look normal, prions may be there somewhere. Only a few misshapen prions might be needed and these can then bind to healthy prions and refold them into disease-causing variants. The biochemical conditions found in brain and neuronal tissue could be important to the onset of infection. Scientists are divided about how disease occurs. The 'Protein-only' theory argues that the misshapen prions are directly responsible for causing disease. Another view is that the misshapen prions get 'outside help' in the form of molecules within the body.

8 What is a neuron?

If only a few mutant prions need to be present, this might support the delay in onset. Many disease agents replicate much faster than infective prions. It takes time for them to refold healthy prions. The mutated products would then go on to reshape healthy ones until there were sufficient numbers to produce a disease condition. This takes longer than replicating genetic material – which prions lack. The misshapen form (PrP) could be only a temporary 'middleman' to infection and not, in itself, the direct cause of infection. Experiments were performed in which brain tissue infected with PrP^{Sc} (which causes scrapie) was injected into the brains of two groups of healthy mice. One group of mice received the PrP^{Sc} combined with the deformed PrP while the other only received pure PrP^{Sc}. Where PrP was absent, infection did not occur. Yet, injecting pure PrP into healthy brains also failed to produce the disease. This suggests that a third piece of the puzzle is missing – but what could it be? Enter 'protein X', the suggested middleman. Some scientists believe that a molecule (probably a protein), must somehow get involved in the whole 'bind and reshape' activity. Laboratory experiments in which pure 'scrapie' prions were mixed with healthy prions (but no other molecules were present), gave interesting results.

The mutant PrP^{Sc} was able to attach to healthy prions (PrP^C) and reshape them but when this happened, the newly deformed (once healthy) prions did not cause infection. They were simply deactivated. Another interesting experiment, using transgenic mice, showed how the different strains might arise based on evolutionary links. *Transgenic* species are those that have genetic material introduced into their genome from another species. A strain of infective prions were extracted from a group of infected hamsters (small rodents) and injected into a group of mice. Although both are related rodent species, infection did not occur. The DNA of the transgenic mice contained some hamster genes and when the same infective prions were injected into this group infection occurred. However, when the prion structure was investigated, it differed slightly to the original material suggesting that the mutant prions are able to 'adapt' and reshape to compensate for genetic differences between related species, thus producing a new strain.

Explain what is meant by the abbreviations:

9 PrP^C

10 PrP

11 PrP^{Sc}

People are willing to accept that cross-infection happens between sheep and cattle because those two species are reasonably closely linked in evolutionary time. People are less willing to believe that, because of the much longer evolutionary distance between humans and these animals, cross-infection can occur by eating infected meat. This will continue to generate debate, though evidence suggests that, eventually, the proof will be forthcoming.

12 Outline the theories on how prions may cause disease.

13 What is meant by a *transgenic* organism?

Research

14 Since this sheet was written, scientists will have discovered new facts about prions and diseases such as BSE and CJD. Outline any new developments not mentioned here.

The human body looks after itself without much thought, or, to quote from one syllabus, 'without any conscious intervention'. There are many things that the human body does without having to think about it – the autonomic system controls many of the involuntary processes that keep us alive.

A dictionary definition of health is 'the condition of being sound in body, mind, or spirit; especially freedom from physical disease or pain'. A disease is anything that impairs (makes worse) how an organism functions. Most diseases are recognisable by what they do to an organism and the symptoms they cause. Doctors often diagnose an illness using the symptoms it causes.

- | | |
|--|---|
| 1 What is meant by <i>intervention</i> ? | 4 Give another definition of <i>disease</i> . |
| 2 Name five things that the body does without thinking to ensure survival. | 5 What is meant by <i>symptom</i> ? |
| 3 Give another definition of <i>health</i> . | 6 Define <i>diagnose</i> . |

Health is maintained in the human body by a range of systems including:

- genes – the units of inheritance. They carry on specific characteristics from generation to generation and can pass on traits (qualities) that help an organism fight off infection.
- mitosis – a form of cell division that results in genetically identical copies of somatic (non sex) cells. This process replaces dead and injured cells. White blood cells (leukocytes) are cells that resist invasion by foreign organisms and are produced by mitosis in large numbers when an infection occurs.
- cell differentiation – when an immature cell (resembling many other cells) changes to a specific mature cell.
- cell specialisation – means that individual cells carry out specific roles; some of these roles involve the maintenance of health.

(Note: Remember the order of increasing complexity: cells / tissues / organs / organ system / body.)

- Name two other characteristics that genes could carry.
- Meiosis is another type of cell replication. Outline how this differs from mitosis.
- Name another type of blood cell apart from white blood cells.
- What is meant by an 'immature' cell?

Diseases

A disease that can be caught from another living creature is described as *infectious*. If it can't be transmitted it is called *non-infectious*. An infectious disease can be caused by:

- | | |
|---|-----------------------------------|
| • prion (proteinaceous + infectious + on) | • fungi (singular <i>fungus</i>) |
| • bacteria (singular <i>bacterium</i>) | • protozoa |
| • viruses | • macro parasites |

- Name ten different diseases that affect humans.
- Give a symptom of each disease.
- Divide your list of diseases from question 11 into infectious and non-infectious.
- Give a definition for *microbe*.

Pathogens

A pathogen is an organism that causes a disease when it lives in or on another organism. Pathogens can cause the following effects when they infect an organism:

- destroy cells
- produce toxins (poisons)
- interfere with the normal functioning of the organism
- cause an immune response (producing antibodies and lymphocytes) that, if excessive, harm the organism.



Causes	Description	Diseases
Prions	Tiny protein particles without nucleic acid, thought to cause various infectious diseases of the nervous system.	scrapie, Creutzfeldt-Jakob disease
Viruses	Sub-microscopic, but complex molecules, that typically contain a protein coat surrounding an RNA or DNA core of genetic material but no semipermeable membrane. Capable of growth and multiplication only in living cells.	chicken pox, hepatitis, influenza, measles, polio, rabies
Bacteria	Microscopic, unicellular organisms.	cholera, pneumonia, tuberculosis
Protozoa	Single-celled parasites, which resemble animals and may form large colonies.	malaria, yellow fever
Fungi	Single or multicelled organisms that produce spores and obtain food directly by absorption. Sometimes classified as plants without chlorophyll.	ringworm, athlete's foot
Macro parasites	Small organisms that live on other organisms and cause them harm, eg ticks, fleas, flukes (a type of flatworm).	schistosomiasis or bilharziasis (a widespread disease caused by the infestation of the human body by flukes)

For a pathogen to infect an organism it must be:

- able to survive the transmission from one host to another.
- able to enter the host.
- virulent – able to overcome body defences and only need to be initially present in small numbers.

One defence barrier to prevent entry of pathogens in humans is skin.

15 Identify three effects a pathogen can cause when it infects an organism.

16 Identify two other external defence barriers against pathogens present in humans.

Antigens

In addition to external barriers such as skin, mucous membranes and cilia, the human body has internal defences against pathogens. These include antigens, molecules that trigger the immune response. Antigens are molecules that invade the body. The immune system recognises them as foreign and produces antibodies to bind to them, thus marking them for destruction. After antibodies have been produced, people are often described as having built up an immunity to a particular illness.

Inflammation is a complex defence mechanism, where the body sends greater amounts of blood and lymph to the affected area. The symptoms of inflammation include swelling, tenderness, increased temperature and redness. Inflammation is a reaction by a part of the body to injury, infection, trauma or environmental conditions.

An injury resulting in tissue damage may be caused by a cut, knock, sprain or repetitive strain injury. Infections are caused by pathogens such as bacteria, fungi and viruses or by the toxins they produce. Trauma can be described as either physical injury or nervous shock experienced by a person. Environmental conditions that cause inflammation may contain allergens (substances that cause allergies) such as animal fur, dust, mites, moulds, pollen, smoke or harsh chemicals.

17 Describe what is meant by *immunity*.

18 Explain what is meant by a *pathogen*.

19 Explain what is meant by a *toxin*.

20 What is *lymph*?

The body's defence system

4

Damaged cells (and white blood cells) secrete increased quantities of histamine, a hormone found in minute quantities throughout the body. Histamine causes blood vessels to dilate, lowers blood pressure and makes the capillary walls more permeable to fluids that need to move out of the small blood vessels to reach the damaged tissue.

- 1 Define *permeable*.
- 2 Define *dilate*.
- 3 Define *secrete*.

The larger diameter of the blood vessels allows more liquid to flow. This in turn causes swelling (due to the increased size of the blood vessels and extra fluid) and tension in stretched skin. The increase in temperature of the inflamed area is due to increased blood flow – blood transfers heat around the body. There may also be an increase in body temperature and pulse rate if a large area is affected. The increased temperature of the body is a defence mechanism to make it harder for pathogens to survive. The swelling may also be painful and throb due to the pulse from connected arteries. Pain is also caused by the release of certain chemicals by the body.

The increased circulation brings special defence agents to the site of cell damage. These include substances to ensure that blood clots in order to stop pathogens from circulating to the rest of the body. *Macrophages* (histiocytes) are specialised types of white blood cells that are attracted to the damaged tissues. They are phagocytes – they surround and digest dead tissue and foreign objects (such as pathogens) that enter the bloodstream. The word comes from the Greek '*phagos*' (one that eats) and '*kytos*' (cell). Phagocytes can pass through the walls of blood vessels.

- 4 The diagram shows a sequence where a macrophage attacks a pathogen. The four stages involved are written below, but in the wrong order. Rearrange the stages in the correct order.

- A The cell membrane of the macrophage forms an indentation.
- B The macrophage approaches the pathogen.
- C The macrophage digests the pathogen and moves on.
- D The macrophage engulfs the pathogen.

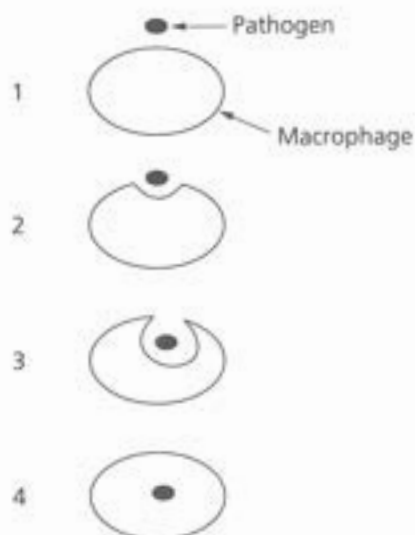
While inflammation is part of the body's defence mechanisms against pathogens, it has some drawbacks. It can damage tissue and cause pain. If too much histamine is produced, especially in allergic reactions, then people can display painful symptoms. Hay fever and some types of asthma can be caused by over-production of histamines and have to be controlled by drugs called 'antihistamines'. If the inflammation response is overdone, then tissue damage can result. Arthritis is a disease where joints are inflamed.

- 5 Construct a table showing the advantages and disadvantages of inflammation.
- 6 Aspirin is often described as an anti-inflammatory drug. What is the likely purpose of anti-inflammatory drugs?

Pain

Pain is an unpleasant experience of the senses that is caused by damage to tissue and detected by sense receptors (special types of nerve cells). These send the message to the brain to tell it that something untoward has happened. As the signal travels from the injury site to the brain, it has to pass across synapses. A synapse is the junction between two neurones, ie the point at which the nervous impulse 'jumps the gap' between them. The brain interprets the signal from the damaged area. Sometimes if pain persists, people take analgesics ('pain killers') to relieve the pain.

- 7 What is another name for a neurone?
- 8 Is pain always a bad thing? Explain your answer.





Cell death can be used to seal off a pathogen. In some cases, a pathogen is completely surrounded by macrophages and lymphocytes so that it is enclosed in a cyst (Greek, bladder, pouch). The white cells doing this die. Because the pathogen is isolated from its food supply, it also dies. Although body cells die in the process, overall the body benefits.

Identify the terms defined below. Choose from *cyst*, *lymph*, *lymphocyte* and *macrophage*.

- 9 White blood cell, typically found in lymph, that makes antibodies.
- 10 A capsule or sac with a distinct membrane formed to enclose foreign material.
- 11 White blood cell that consumes foreign material and protects against disease.
- 12 Blood that doesn't have red blood cells, platelets or large plasma proteins.

Immunity against disease

The components of this part of the body's defence against infection include:

- **Antibodies** (immunoglobulin): protein produced in lymphocytes as a result of presence of antigens.
- **B-cells** (bone-marrow-derived cell): lymphocytes that secrete antibodies.
- **T-cells**: lymphocyte originating from the thymus gland that reacts with antigens.

Helper T-cells activate killer T-cells and also B-cells, by secreting chemicals that stimulate cloning in T-cells and B-cells. Suppressor T-cells (which only live for a short time) suppress T-cells and B-cells when antigens have been destroyed and there is no longer a need for as many lymphocytes. Memory T-cells remain and when an infection occurs by the same antigen they are reactivated quickly. Helper T-cells stimulate B-cells and T-cells to replicate. In theory, a T-cell could attack a B-cell that had an antigen on its surface. In practice this doesn't normally occur because T-cells are able to recognise individual surface molecules of the millions of different B-cells – just like having a 'fingerprint'. Only genetically identical individuals will have the same surface molecules on their B-cells.

- 13 Briefly describe, with reference to genetic properties, the types of cells produced by cloning.
- 14 What is an *antigen*?
- 15 Give an example of genetically identical humans.
- 16 Describe the role of helper T-cells.
- 17 Where do T-cells originate?

Major Histocompatible Complex (MHC)

The body's immune system needs to be able to identify 'self' cells from 'non-self'. With the exception of identical twins, the proteins used for such identification are unique to each individual. *Major histocompatible complex molecules* (MHCs) are protein-based substances that associate with lymphocytes, the T-cells in particular. Their name is derived thus:

- *Major* – important; *Histo* – to do with histamine responses (eg allergic reactions); *Compatible* – well matched, in harmony with something; *Complex* – mixture, compound

The outer membrane of every cell within our body has recognisable MHC markers projecting from it. MHCs act like a 'braille' system for the immune system, able to bind to and identify non-self proteins within a body, by reading the markers. This is why organ transplants can fail. The MHCs quickly recognise the proteins found on the transplanted tissue as non-self and set out to destroy it. The closer the match, coupled with immuno-suppressing drugs (which slow down or stop immune responses), the more likely it is that a transplant will be successful. When the lymphocytes identify the MHC marker, they release the cell and move on. However, if the cell contains any foreign matter (perhaps a bacteria that has attached itself to the cell), despite having the marker, it is targeted for destruction.

- 18 Suggest the dangers faced by people undergoing immuno-suppressing medications.

The role of white blood cells

There are five main types of white blood cells or *leukocytes* (*leuko* white/colourless, *cyte* cell). All leukocytes are formed from stem cells present in bone marrow. Depending on the cell type, the leukocyte will either mature while still in the bone marrow or, as in the case of the T-cells, they are formed in the bone marrow then travel to the thymus to mature. (The thymus is situated near the breastbone/tracheal region.)

There are fewer white blood cells in the blood than red blood cells. It is estimated that they are outnumbered by at least 700 to one. However, this does not mean they are found in small numbers! In fact, a conservative estimate would number hundreds of millions (probably more) though many die within 24 hours. Once formed, white blood cells begin circulating throughout blood, the lymph system and tissues.

When body tissue is damaged, *chemicals* are released that attract white blood cells to the site of the damage. Invading bacteria also cause the release of chemicals that attract white blood cells. The primary role of all leukocytes is to protect the body from invasion by disease-causing viruses and bacteria. White blood cells also perform general housekeeping duties by digesting dead and dying cells and other unwanted matter found within the body. The three main types of white blood cells are: *granulocytes*, *monocytes* and *lymphocytes*.

1 Give a definition for stem cells.

Up to 75% of the body's white blood cells are called *granulocytes* (*granum* grain) with estimates of 50–60% on average. Granulocytes get their name from the grainy substances found in the cytoplasm. Therefore, if it is a white blood cell with the grainy cytoplasm, regardless of the role it plays, it is classified as a granulocyte. There are several types of granulocytes and each is identified by the way it stains. These are:

- *Basophils*, which are identified by staining a pale colour. They are associated with antihistamine reactions, though this does not mean they perform no other role.
- *Neutrophils* work with the immune system's 'search and destroy' cells. They stain a neutral colour.
- *Eosinophils* play a major role in allergic reactions. They have also been identified working with other leukocytes, particularly with assisting chemical reactions. Eosinophils stain a reddish-brown colour and are believed to represent the smallest group of leukocytes with estimates at less than 5% of the total.

2 Give two reasons why scientists might stain cells.

As a whole, the different granulocytes play varying roles in the body's defence system and might therefore be considered as a group of 'all rounders'.

Monocytes are large cells known by two different names, depending on where they are found within the body at the time. They are called monocytes when circulating (and working) throughout the blood system. However, not all substances that must be dealt with are found in the bloodstream. At times, the monocyte is called to digest something (be it an invader or one of the body's own dead cells) that is found within tissues or other areas. When it leaves the bloodstream to pursue a target, it is then called a *macrophage* (*macro* large, *phage* to eat/digest). When it returns to the blood, it is again referred to as a monocyte. Some sources will state that the monocyte evolves into a macrophage but it is better to consider their dual roles as described above. Approximately 7–9% of leukocytes are believed to be monocytes.

3 Which of the white blood cell types is the most abundant?

Antibodies and antigens

Not all white blood cells fight disease by digesting invaders. The *lymphocytes* do their bit by converting globulins (proteins found in blood plasma)* into antibodies. Antibodies are molecules that bind to invading substances thus 'tagging' them for destruction by the macrophages. The targeted substances are known as *antigens*. An antigen is often recognised as a protein found on a bacteria or virus 'outer coat' and it causes the production of an antibody. Depending on the type, antibodies can destroy or inactivate bacteria and viruses or bind to the antigen and 'flag' it for the macrophage to deal with.

* There are a number of different globulins found within the blood plasma.



Antibodies are very specific. Once a person has had a particular infection the antibodies produced will stay dormant in the body but will be reactivated whenever that particular antigen reappears. This gives a certain amount of immunity to subsequent infections from the same type of microbes. It also saves time because the immune system can act faster if it has already dealt with the antigen.

- 4 What is blood plasma?
- 5 How is the name *macrophage* derived?
- 6 Give a definition for a *globulin*.
- 7 Define *dormant*.
- 8 Explain the difference between a *monocyte* and a *macrophage*.

B-cells and T-cells

There are two main sub-groups of lymphocyte: B-cells and T-cells. The *B-cells* are responsible for antibody production. When an invader is recognised, antibodies are produced from proteins present in blood plasma. The antibodies then attach to the non-self invaders, flagging them for destruction. Like all lymphocytes, B-cells are formed in the bone marrow and this is where they mature. In other words, once they leave, they are ready to work.

There are three types of *T-cells*. As the name suggests, the *Killer T-cells* are the 'search and destroy cells'. These cells destroy invaders and dead or damaged 'self' cells that are recognised as unfit – including some forms of cancer cells. While other forms of treatment are usually required to fight cancers, Killer T-cells abound where it is present. T-cells form in the bone marrow but must migrate to the thymus to complete maturation. Once mature, many will head for the lymph tissue while others will circulate throughout the bloodstream.

- 9 The lymph glands are positioned at strategic parts of the body to enable a quick response by the glands closest to infection. Suggest how storing a number of T-cells in lymph tissue at all times is of benefit to fighting disease.

Two 'brothers' to the Killer T-cell are the *Helper T-cell* and the *Suppressor T-cell*. Helper T-cells activate the required immune system responses to warn the Killer T-cells and the B-cells that trouble is brewing ('Wake up! We're being invaded!'). The Helper T-cells cause the existing population of Killer T-cells and B-cells to rapidly and repeatedly divide, thus creating a cloned army that will set out to attack the invaders. Later, as the activity winds down and the 'army' is no longer needed, the Suppressor T-cells slow down the activity that the original immune response activated (*suppress* to restrain or slow down).

- 10 Suggest what might happen if the Suppressor T-cells did not slow down the Killer T-cells and the B-cells after the invading organisms had been dealt with.
- 11 What does the following describe: 'A molecule that initiates an antibody reaction.'?
- 12 Summarise the roles of B-cells and T-cells.
- 13 Describe the difference between an *antibody* and an *antigen*.
- 14 **Mini project!** Draw a small, cartoon-style picture illustrating the roles of the different types of white blood cell.

Did you know? The gooey substance we call *pus* is made up of dead tissue and dead bacteria but mostly dead leukocytes – most of which are neutrophils.

Pathogens and antigens review quiz

Pathogens can enter the body by several routes: nose, mouth, skin and reproductive organs. *Pathogens* are microorganisms that cause disease. *Antigens* are molecules that trigger the immune response when pathogens are detected in the body.

Once inside the body there is often a gap in time between infection and the appearance of symptoms associated with it. This is known as the *incubation period*. During this time the invading microorganisms rapidly increase in numbers. Some pathogens release poisonous substances called *toxins*. Louis Pasteur and Robert Koch were the first people to prove that diseases were caused by microorganisms. Pasteur also showed that some microorganisms could be killed by heat.

B- and T-lymphocytes: assumed knowledge quiz

1 Study the description of 'immune response' below and give definitions of the terms in *italics*.

Some white blood cells respond to the presence of particular foreign substances (antigens) in the bloodstream. An antigen is often a protein found in bacteria or a virus, which causes the production of an antibody. These antibodies can destroy or inactivate bacteria and viruses. Antibodies are very specific. Once a person has had a particular infection, the antibodies produced to fight it will stay *dormant* in the body but will be reactivated if the same antigen reappears. This gives a certain amount of *immunity* to subsequent infections from the same type of *microbes*.

The immune response occurs when the body reacts to the presence of an *antigen*. When *lymphocytes* identify the antigenic molecule as foreign, and induce the formation of *antibodies*, together they react to fight the invader thus rendering it harmless to the body.

2 Name three components of the 'third line of defence'. _____

3 Where are all white blood cells formed? _____

4 B-cells can differentiate into either plasma cells or _____ cells.

5 Antibodies and antigens form complexes that agglutinate. Define *agglutinate*. _____

6 IgM, IgG, IgA, IgD and IgE are all forms of _____.

7 What purpose do memory B-cells serve? _____

8 In which part of the body do T-cells mature? _____

9 T-cells can secrete a chemical called perforin. Describe the function of this substance. _____

10 Which cells provide 'cell-mediated immunity'? _____

11 Give another name for T4 cells. _____

12 Give another name for T8 cells. _____

Indicate whether the following statements are true or false. _____

13 Granulocytes represent the largest group of white blood cells. _____

14 Antigens are 'invading' substances. _____

15 Macrophages are not the same type of cell as monocytes. _____

16 There are more white blood cells in blood than red blood cells. _____

17 T-cells exist in three forms. _____

18 There are five main types of leukocyte. _____

19 Globulins are a type of blood protein. _____

20 The most abundant white blood cell type is the lymphocyte. _____

21 Suppressor T-cells act to slow immune response activity. _____

22 The collective name for all white blood cells is leukocytes. _____

23 Antibodies are protein-based substances. _____

24 White blood cells are formed in bone marrow. _____

25 T-cells mature in bone marrow. _____

26 Granulocytes are identified by staining their cytoplasmic 'grains'. _____

Robert Koch

Robert Koch (1843–1910) was a German scientist who won the 1905 Nobel Prize for Medicine. He is regarded as the co-founder (with Louis Pasteur) of modern microbiology. Koch is regarded as one of the greatest bacteriologists the world has ever known. He had a wide range of interests, especially bacteriology, a science that was then in its early stages. Koch's first major finding in bacteriology was in 1876, when he demonstrated that if viable *Bacillus anthracis* was injected into healthy mice, the mice eventually developed anthrax.

While many of the causes of disease are now recognised, in the past people did not realise (or refused to believe) that tiny organisms could be responsible for certain illnesses. Koch was the first person to show conclusively that the cause of an infectious disease was a specific organism. He was also able to identify many of the organisms associated with particular diseases.

- 1 Define *viable*.
- 2 Suggest why people may not have believed that microorganisms were responsible for disease.

Koch was one of the first scientists to obtain microbes from infected animals and show how these microbes could be grown in artificial conditions. The techniques he developed were used to isolate many different types of bacteria. He also demonstrated the importance of being able to destroy the cultured microbes safely after research had been carried out. Koch was one of the first people to advocate the sterilising of surgical instruments with steam. Louis Pasteur also promoted the importance of sterilising surgical instruments, bedding and other items used in hospitals.

- 3 Define *sterilising*.
- 4 Give an example of when bedding might need to be sterilised.
- 5 The terms *sterilisation* and *disinfection* are often interchanged. While they are similar in many respects, there is a fine difference between them. Explain the difference between the two terms.
- 6 Suggest the conditions that would need to be met to grow microbes artificially.
- 7 Explain why we might want to grow microbes, even if they are harmful.

Tuberculosis (TB) is an infectious disease of the lungs spread by airborne droplets from the sputum of infected people. Estimates vary, but between a quarter and a third of all people have the bacteria causing TB present in their body. The disease tends to be most common where there is overcrowding, poor nutrition and poor health care. It currently kills over two million people a year. In Koch's time it was almost invariably fatal. The treatment was for the patient to be cared for in a sanatorium until they died. Koch worked for many years on isolating and cultivating bacteria. He was highly regarded by his contemporaries. One was so impressed by Koch's work on the bacteria causing anthrax that he stated "This young Robert Koch will surprise and shame us by the brilliance of his investigations". In 1882 Koch isolated the bacteria (*Mycobacterium tuberculosis*) causing the disease. He developed the tuberculin test in 1890 which helped doctors better diagnose the disease.

Eventually, in 1924, the BCG vaccine for the disease was developed by other doctors. In more recent times antibiotics have been used to treat tuberculosis but the disease has developed a resistance to some of these. It has also become a problem for people whose immune system is not at full strength. One survey in the USA reported that people working in nursing homes are three times more likely to develop TB than similar individuals with different occupations. Domesticated animals such as cattle, chickens and pigs can carry the disease but it is rare in wild animals.

X-rays can show development of TB in the chest but more specific tests are needed to confirm the disease.

- 8 What is a sanatorium?
- 9 What is meant by the phrase: 'It was almost invariably fatal'?
- 10 Is having the presence of TB bacteria present in your body a definite indication that you will catch the disease?
- 11 Give a reason why more people working in nursing homes might exhibit symptoms of the disease than is generally expected.
- 12 Give two techniques that can be used to screen for TB.



Research

13 Find out what the initials *BCG* represent.

14 Summarise the symptoms of tuberculosis. Include a description of how it is spread.

Koch's postulates

In 1890 Robert Koch set out his criteria for judging whether a given bacterium was the cause of a given disease.

- The same bacteria must be found in all cases of a given disease.
- The bacteria must be isolated from the diseased host and grown in pure culture.
- The isolated bacteria must reproduce the same disease when inoculated into a healthy susceptible animal.
- The bacteria must be recoverable from the experimentally infected host.

Limitations of Koch's postulates

- A bacterium that is a normal part of the environment may become a pathogen in certain situations. They can mutate and acquire extra virulence factors, making them pathogenic. Some usually harmless bacteria can cause disease if they gain access to deep tissues by trauma or surgery.
- In immunocompromised patients, many free-living bacteria and components of the normal flora can cause disease, especially if they reach deep tissues.
- Not all of those infected or colonised by a pathogenic bacterium will develop disease. Subclinical infection is usually more common than clinically obvious infection.
- Some bacteria are not culturable *in vitro* (outside the body and in an artificial environment).

15 What does it mean when a person is *immunocompromised*?

16 Organisms grown *in vivo* are grown inside another organism. Describe how organisms are grown *in vitro*.

17 Explain what is meant when an animal is *susceptible* to a disease.

18 What conditions might cause an animal be more likely to catch a disease and why?

19 Suggest how a mutated form of a usually harmless microorganism might suddenly become dangerous.

20 Discuss why, if Koch's postulates have limitations, they are still used by medical researchers.

21 Suggest why some species of bacteria cannot be cultured *in vitro*.

Research

22 Koch is famous for his work with five of the diseases listed below. Identify the five diseases.

Anthrax, cholera, Ebola, leukaemia, malaria, sleeping sickness, syphilis, tuberculosis

Malaria

Insects can help or harm humans. Bees pollinate plants used for food but many other insects can harm human health. When acting as disease-causing vectors, insects are harmful to humans and other animals. The female anopheles mosquito feeds on blood (unlike the male counterpart that feeds on nectar), and is probably the insect that has caused the most death and suffering in human history. Female mosquitoes need blood to produce eggs. In taking blood, the mosquito can pass on malaria to a person. This is called the *mechanical transfer* of an infective agent. In this case, females inject sporozoites, a motile infective form of parasitic protozoa that reproduce asexually in the new host.

- 1 What is *nectar*?
- 2 What is a vector?

Malaria is one of the world's most deadly diseases and kills millions of people every year. The mosquito does not directly cause malaria, but it infects humans with a protozoan parasite – *plasmodium* – that it carries within its own body. There are four species of malaria-causing plasmodium and one of them will be present in an infected person. The plasmodium finds its way into the mosquito when it bites someone infected with malaria. The parasite flows into the mosquito along with the blood that is drawn up from the bite. It will then reach a certain phase of development within the mosquito, after which it migrates to the salivary gland. There it mixes with saliva and anticoagulant used to 'thin' the blood of the host organism when it is bitten.

A female anopheles mosquito pierces the skin with a sharp mouthpart and injects saliva into the wound. Blood is sucked up by the mosquito's proboscis (tube). If the mosquito is infected with the protozoan parasite, then bites another human, the parasite will be injected into the person's bloodstream. Once there, the parasite migrates to the liver where it matures, replicates and causes a full-blown attack by infecting the red blood cells. Mosquitoes are *ectoparasites* – they live on the exterior of the host.

- 3 Define *parasite*.
- 4 Define *motile*.
- 5 Explain why a mosquito might benefit from injecting an anti-clotting agent into a person.

Protozoa are unicellular organisms. *Plasmodia* replicate in the liver for 2–4 weeks and then invade red blood cells, which eventually burst and release even larger numbers of plasmodia. Lysis, coupled with substances produced by the immune system, lead to the onset of fever and chills. The disease can be broken down to four stages. When the red blood cells have been infected to the point where there are so many spores within them that they burst, a malaria 'attack' occurs. Once infected with malaria, recurring attacks are not uncommon, particularly during times of stress. Symptoms will depend on the species of malaria involved but they include a combination of high fever and chills.

Identify the four stages of malaria below (6–9) using the terms *fever*, *incubation*, *shivering*, *sweating*.

- 6 Protozoa multiply but no symptoms are shown for the first one to two weeks.
- 7 Sufferers have an elevated body temperature, but feel cold for a short period.
- 8 Lasts for several hours, body temperature can rise above 41°C and red blood cells are bursting.
- 9 After the fever, the sufferer can sweat for several hours and usually recovers (about 1% of cases are fatal).
- 10 Without medical treatment the attacks return, even after person seems to have recovered.

Name a drug used to treat malaria.

The name *malaria* comes from the Italian words *male*, meaning 'bad', and *aria* which means 'air'. Originally people thought malaria was caused by 'bad air'. Ronald Ross (1857–1932) was a British surgeon who studied malaria in India and came to the conclusion that malaria was carried by an insect vector – the female anopheles mosquito. Ross unravelled the complicated life cycle of the plasmodium parasite. This parasite bred in the stomach of the female mosquito, and then was carried by the bloodstream to the mosquito's salivary glands. When a victim was bitten the parasite was carried with the mosquito's saliva into the victim's bloodstream where it reached maturity. His work earned him the 1902 Nobel Prize for Medicine.



11 Suggest why malaria is prominent in tropical and subtropical areas as opposed to other parts of the globe.

Mechanical transfer is not the only way insects can infect other animals. *Biological transfer* is a less random method. It involves a three-step process and cannot succeed if any step is missing or interfered with. The availability of the correct insect vector is the first step. The pathogen (disease-causing agent) must first replicate to sufficient numbers and reach a certain stage of development within the vector. The vector does not, however, have all of the necessary features for full infection to take place. Therefore, once sufficient numbers of the pathogen are prepared, it requires a host organism before full-blown infection is complete. The insect carries the pathogen to a host organism that provides what is required. The host organism is the 'final stop' and the one where maturation is complete. Different species of mosquito carry different forms of plasmodium and the combinations determine which animals will be affected by malaria. An estimated 125 million new cases of malaria are reported every year.

12 Why do female anopheles mosquitoes bite humans?

13 What is a *proboscis*?

14 In addition to transmitting protozoa, insects can infect animals with viruses and nematodes.
What is a *nematode*?

15 Describe the difference between *mechanical transfer* and *biological transfer* of infective agents.

16 Pathogens spread by biological transfer usually require specific vectors. With this in mind, what would be the best way to reduce the rate of infection by these pathogens.

Cure and prevention

There have been many ways used to cure the symptoms of malaria or to prevent the disease. Quinine was originally extracted from the bark of a tree, but is now synthesised, and can relieve the symptoms of sweating and fever. Insect repellent is useful in ensuring that insects don't bite and hence can't infect. Netting or mesh over windows and beds can stop mosquitoes from reaching a human to bite them. Removing areas of stagnant water can stop insects breeding and insecticides can kill mosquitoes.

17 Put the information on curing and preventing malaria into the table below.

Method	Advantage	Disadvantage

Research

18 Malaria is not the only important disease transmitted by flying insects. Other examples include: sleeping sickness (trypanosomiasis), dengue fever, yellow fever and river-blindness (onchocerciasis). Investigate the causes and symptoms of one of the examples given above. Write a brief summary describing your choice.

19 People with *sickle cell anaemia* have a greater resistance to malaria than those who do not carry the sickle cell gene. Give brief details for sickle cell anaemia.

Respiratory diseases

The body uses various strategies to fight disease but sometimes illnesses arise that leave it, and the scientific world, dealing with exceptions. *Pneumonia* is a respiratory condition that causes an inflammation of lung tissue. The tissue firms up and fluid builds in the lungs, particularly in the alveoli. This causes problems with oxygen intake and the reduced oxygen levels may need to be supplemented using medical techniques such as the oxygen mask, which provides pure oxygen for the patient to inhale. Certain strains of bacteria and viruses are associated with typical pneumonia. There are around 50 types of pneumonia.

- 1 Outline the differences between bacteria and viruses.
- 2 Pneumonia caused by bacteria (eg *Streptococcus pneumoniae*) can be treated by antibiotics such as penicillin. Pneumonia caused by a virus cannot be treated with antibiotics. Explain why this is so.

Influenza is a highly contagious disease of the respiratory system caused by various strains of viruses. The symptoms include a fever, tiredness and aches and pains. It can occur almost anywhere in the world. The incubation period for influenza is 1 to 3 days and the illness lasts about 10 days.

- 3 What treatment is usually advised for someone suffering from flu?

Regularly new strains of influenza appear, many of which have been traced back to originating in China. When the disease affects large parts of the community, an *epidemic* can arise. Should the disease spread to other countries, then a *pandemic* results (*pan* all). In 1918/19 an outbreak of influenza (the 'Spanish flu') claimed 40 million lives. Soldiers returning from World War I in Europe are thought to have brought the disease to Australia, resulting in 12 000 deaths. Other large-scale infections have included the 'Asian flu' (1957) and the 'Hong Kong flu' (1968). Between 1995 and 2002, there were several outbreaks of influenza in Hong Kong and millions of chickens were inoculated or slaughtered as a result.

- 4 Give reasons why the authorities got rid of so many chickens at that time.
- 5 Outline the difference between an epidemic and a pandemic.

Atypical pneumonia: Severe Acute Respiratory Syndrome (SARS)

Severe, when referring to a disease, means that the symptoms are extreme or dangerous. An *acute* disease is one in which symptoms arise quickly and don't last long. When the signs are present to indicate that a patient has pneumonia, but the illness is not following the typical pattern of development, it is called *atypical pneumonia* (*atypical* unusual). In March 2003 the media informed the public about the appearance of an atypical pneumonia called *severe acute respiratory syndrome*, or SARS. The SARS outbreak was traced back to Guangdong Province in China. It first appeared there in November 2002 but the authorities did not make it known and, within months, people visiting China had exported it to over a dozen countries. By late April 2003 SARS had reached 27 countries leaving 5000 affected and about 300 dead.

SARS was first believed to be a variant of the influenza virus but its incubation period is slightly longer – about 2 to 7 days (occasionally up to 10 days). While influenza has not been completely ruled out, other paths are being investigated, as researchers discovered a previously unrecognised *coronavirus* in patients with SARS.

- 6 Give a reason why authorities might be reluctant to report cases of a potentially fatal disease.
- 7 Give your own definition for *severe*.
- 8 Give your own definition for *acute*.
- 9 What does '*SARS was believed to be a variant strain*' mean?

The coronavirus gets its name from having a halo, or crown, around the virus particle (*corona* crown). The crown is formed by projections coming from the virion's outer coat and this is a distinctly recognisable feature of the *Coronaviridae* family. *Coronaviridae* contain a single strand of RNA and have different forms during the various stages of their lifecycle – this is called being *pleomorphic*. They are linked to respiratory illness, hepatitis (inflamed liver) and gastrointestinal infection. The condition caused by the virus, and its severity, depends on the animal species affected.



Although previously unknown coronavirus has been isolated in SARS patients, however this does not mean that it is the main cause of the illness. Another RNA virus, the *paramyxovirus*, is also being investigated for possible links to SARS. Paramyxoviruses have a helical shape and the virus affects vertebrates.

Symptoms of SARS

Many of the symptoms associated with SARS might first be mistaken for flu. It begins with a fever of 38°C (104°F) or greater. Some people might also develop chills. Within 48 hours of the onset of fever, a dry cough develops. This leads to breathing difficulties and should not be ignored. Headache, muscle stiffness, loss of appetite and other flu-like symptoms can also develop but the fever and dry cough are indications that it could be more than flu. If the patient has also recently been overseas (particularly to Asia) or recently associated with someone who has, this information must be given to the doctor involved.

10 Name another illness with symptoms that might be mistaken for flu.

11 Suggest why so many illnesses share similar symptoms.

12 Viruses are often referred to as 'non-cellular'. Explain why.

Since viruses do not respond to antibiotics, there is no single cure for SARS. Doctors must treat the symptoms instead. These include giving antibiotics for the types of bacteria known to accompany typical pneumonia, as well as antiviral agents and whatever else the doctor feels will help to relieve the symptoms. At present, *droplet infection* (particles sneezed or coughed out) and contaminated objects (such as utensils) are considered the most likely cause of its spread.

13 Why do viruses not respond to antibiotics but bacterial agents do?

14 Since viruses do not respond to antibiotics, why would a doctor prescribe them to a SARS patient?

Many people get a vaccination for influenza every year just before the flu season begins. They can be those who have a greater risk of contracting influenza or who might suffer complications if they catch the disease. The vaccine used has to change at regular intervals because the strains of viruses that cause influenza can change rapidly. Common advice is to stay at home, rest, drink plenty of fluids and take aspirin or paracetamol.

15 Identify two groups of people who stand a greater than average chance of catching influenza.

16 Identify two groups of people for whom catching influenza might pose a great health risk.

17 Give two reasons why a person with influenza might be advised to stay at home.

18 A person who had an effective vaccination against influenza might be totally unprotected a year later. Explain why this is.

Research

19 Explain the difference between a *virus* and a *viroid*.

20 Since this article was written, scientists will have learned much more about SARS. Outline any facts about SARS that are not mentioned here. (Hint: for more information about this, or any other illnesses that are globally spread, the *World Health Organization (WHO)* is a good place to start your search.)

Louis Pasteur and Sir Frank Macfarlane Burnet

Louis Pasteur

Louis Pasteur (1822–1895) was a pioneer in the search for the causes of disease in the second half of the 19th century. His work included:

- how to kill bacteria in wine and beer without spoiling the flavour of the beverage
- disproving theory of 'spontaneous generation'
- discovery of bacilli causing disease in silkworms
- discovery of inoculation against the bacteria that cause anthrax.

1 Define beverage.

Louis Pasteur was the founder of the science we now call *microbiology*. Microbiology is the study of microscopic forms of life. In 1854 Pasteur moved to Lille, France, to take up a post at the local university. One of his projects was to investigate why some beer and wine went sour – a problem that cost the French manufacturers of these products a great deal of money. The manufacture of both beer and wine involves the changing of carbohydrates into ethanol (alcohol) in a process called fermentation.

Pasteur examined many fermenting mixtures under the microscope and found different kinds of microbes. He demonstrated that yeast was responsible for forming ethanol and that other microorganisms (bacteria) caused the production of ethanoic (acetic) acid – the substance causing the sour taste. Vinegar, which is mainly ethanoic acid, is named from the French '*vinaigre*' meaning 'sour wine'.

2 The bacteria causing ethanol to change to ethanoic acid are *Acetobacter*, a type of anaerobic bacteria. What is meant when bacteria are anaerobic?

3 What is the chemical name for vinegar?

Pasteur stated that fermentation and decay were both caused by microorganisms and if these microorganisms could be killed after fermentation had occurred in wine, the wine would not go sour. His first attempts to do this involved boiling the wine, which, although it killed the microbes, also spoiled the flavour. After many experiments, he found that a good compromise was to heat the wine to around 60°C for a short period of time. This process (pasteurisation) is still used to kill bacteria in milk and beer but small amounts of sulfur dioxide are now added to wine to act as a preservative instead of heating it.

4 Does pasteurisation kill all of the microbes in milk? Explain your answer.

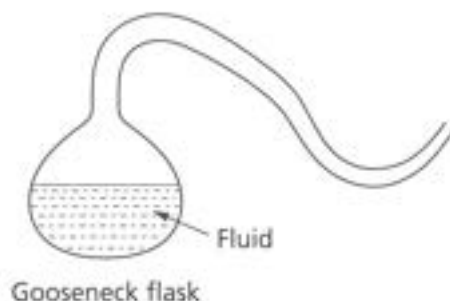
Some scientists in Pasteur's time carried out experiments that involved boiling hay with water under sterile conditions and these, if left for a few days, generated large numbers of microorganisms. They used these results to support the theory of spontaneous generation of life. Louis Pasteur carried out similar experiments using sugar and yeast mixtures boiled in water. He found that only those mixtures exposed to the air grew microorganisms.

5 Explain what is meant by *sterile conditions*.

6 Compare the difference between an *aerobic* organism and an *anaerobic* organism.

7 What sort of organism was found in the boiled hay water – aerobic or anaerobic? Explain how you came to this answer.

In 1864 Pasteur gave a demonstration to the French Academy of Science. He prepared 60 glass flasks containing infusions (mixtures) of water, yeast and sugar. The contents were boiled for two minutes to kill anything present and all but four of the flasks were sealed. These four flasks had the necks melted by heat and drawn out into a long curve, with the necks remaining open to the atmosphere. Pasteur then boiled the contents and let steam come out of the neck of each flask in order to kill any germs present.





The remaining flasks were then split into three groups. The first group was opened briefly and then resealed near where the scientists were holding their meeting. The second group was opened and resealed high in the same building. The third group was opened and resealed outside the building. The flasks were left for a few days.

The four flasks with open, curved glass necks (gooseneck flask) remained sterile. All other groups had some flasks with microorganisms growing and some remaining sterile. Pasteur's hypothesis was that microbes from the air needed to reach the contents of the flask before microorganisms could grow, and microbes from the air entering the curved necks of the open flasks could not travel through to the nutrients.

- 8 Why were the contents of the flask boiled?
- 9 Explain the difference between fermentation and decay.
- 10 What is an infusion? Give an example.
- 11 Suggest why the groups of flasks described above were exposed to air under different conditions.

Anthrax and Pasteur

Anthrax is a contagious disease of cattle and sheep (and other warm-blooded animals, including humans). In Pasteur's time, it caused the deaths of large numbers of livestock and occasionally humans. The cause of the disease is *Bacillus anthracis*. Pasteur was not the first person to develop a culture for this type of bacteria (Robert Koch was the first, in 1863), but he was the first person to develop a vaccine for anthrax.

Pasteur developed a method of inoculating with attenuated (weakened) cultures of germs to give immunity against disease. He publicly demonstrated, with a flock of sheep, that vaccinated animals survived a subsequent infection of anthrax and unvaccinated animals had an almost 100% mortality rate when infected with anthrax.

Pasteur had to work hard to overcome the prejudices of many doctors and scientists of his time, since it meant proving what many of them believed to be wrong. They could not believe that tiny organisms such as bacteria could kill people, but thought that the role that germs played (if they existed at all) was not of any great importance. Pasteur played a major part in the gradual acceptance by the medical community of the fact that diseases could be caused by bacteria.

- 12 Define *culture* in scientific terms.
- 13 Define *vaccine*.
- 14 What is meant by 'an almost 100% mortality rate'?
- 15 What is meant by 'immunity against disease'?

Pasteur was one of the main proponents of the 'germ theory of disease', the theory that diseases are caused by the body being attacked by microorganisms. People found it hard to believe that organisms, many of which could not be seen by the naked eye, could kill people. This was also contradictory to the generally held belief of the time in '*spontaneous generation*' of life. At the time, many people thought that simpler forms of life, such as insects, could arise from inorganic substances under certain circumstances – '*abiogenesis*'.

- 16 Which invention helped to disprove *abiogenesis*?
- 17 What does this define: '*The use of heat to kill pathogens in a liquid without causing major changes to its properties, eg taste and appearance*'?
- 18 Name two liquids that are pasteurised.
- 19 Which organism is used to ferment sugars to make beer and wines?
- 20 Name another food that is prepared using fermentation.
- 21 Which disease is caused by *Bacillus anthracis*?

- 
- 22 One of Pasteur's favourite quotations was: 'Where observation is concerned, chance favours only the prepared mind.' Rewrite Pasteur's quotation in your own words.
- 23 Explain how a gooseneck flask can stop microorganisms from the air coming into contact with the contents.
- 24 Vaccines can be formed using dead strains of an infective agent then injecting the dead organism into the body. How might this be of use in helping to protect an animal from the live, infective form?
- 25 With reference to question 24, why might it be impossible to inject small amounts of certain *live* infective agents into a body instead? Include an example of an infective agent that could not be injected in its live form.

Research

- 26 Who was the first person to see microorganisms?
- 27 Outline Pasteur's work in the discovery of bacteria causing disease in silkworms.
- 28 Pasteur was not the first person to inoculate against disease. This was thought to have been William Jenner in around 1798. Give an outline of Jenner's work in vaccination.

Sir Frank Macfarlane Burnet

Sir Frank Macfarlane Burnet (1899–1985) was an Australian (born in Traralgon, Victoria) *virologist*. A graduate from the University of Melbourne, he worked as a doctor, *pathologist* and ship's surgeon. He gained a doctorate (PhD) in London and, from 1928, became the Assistant Director of the Institute of Medical Research. By 1944 Sir Frank was the Director of Experimental Medicine at University of Melbourne (1946–1965).

He made a great contribution to our knowledge of viruses. He (with Peter Medawar) was awarded the 1960 Nobel Prize for *Physiology* for discovery of acquired *immunological* tolerance to tissue transplants. Burnet's predictions concerning the ability of the body to accept transplanted tissue were confirmed in *graftings* conducted by Medawar on embryo rats inoculated with foreign tissue.

When he returned from the UK in 1928, Burnet studied *bacteriophages* and later turned his attention to viruses. He did important work on the use of chicken *embryos* for the isolation and analysis of animal viruses. These embryos were used to develop *vaccines*. His major work was the *clonal* selection of antibody production. This theory helped to explain how the body could tell the difference between entities as either 'self' or 'non-self'. Sir Frank discovered that antibodies were not *inherited*, but were produced during foetal development.

- 29 Write the meanings of the words in *italics* in the passage above.
- 30 Describe what *acquired immunological tolerance* means.
- 31 Suggest how the chicken embryos mentioned above might be used to develop a vaccine.
- 32 Why might a scientist use embryos instead of fully grown specimens?
- 33 Give an example of a 'self' cell and a 'non-self' cell.
- 34 Briefly describe how a bacteriophage differs from a virus.
- 35 Why is it important for tissue transplants to be tolerated by the immune system?
- 36 We all know that humans are animals yet people often draw an invisible dividing line between 'human animals' and other animals. Nonetheless, humans are often infected by diseases of other animals. Give an example of a disease humans can catch from other animals. Include its symptoms, how it is contracted and possible treatment.

Research

- 37 Give the name and details of another scientist who has won a Nobel Prize for Physiology.

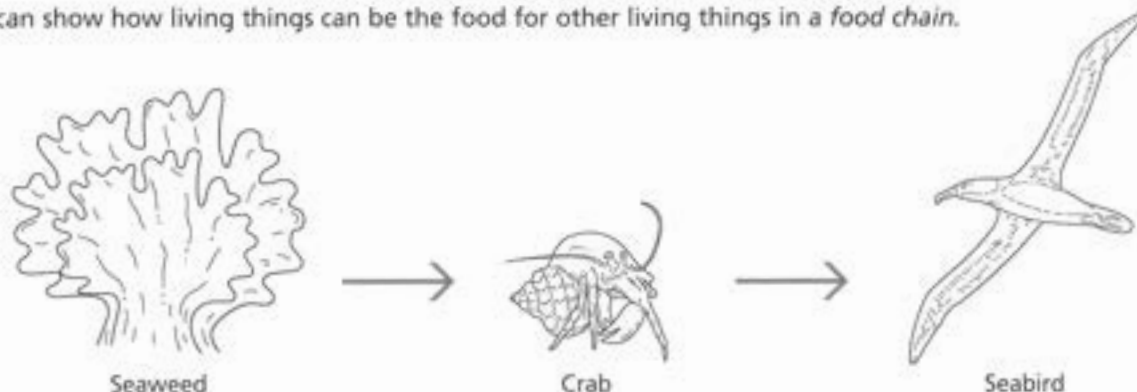
Dependence among organisms

All living things need energy to grow, move, reproduce etc. This energy is provided by food. Plants make their own food using the energy from sunlight in a process called *photosynthesis*.

- 1 Name one use of energy by organisms not already mentioned.
- 2 What is the type of energy that comes from the sun?

Green plants are called *producers* because they make their own food. Animals cannot make their own food. They get their energy by eating (consuming) plants or other animals. If there were no plants to eat, animals could not survive. We also call organisms that produce their own food *autotrophs* and those that do not produce their own food are called *heterotrophs*.

We can show how living things can be the food for other living things in a *food chain*.

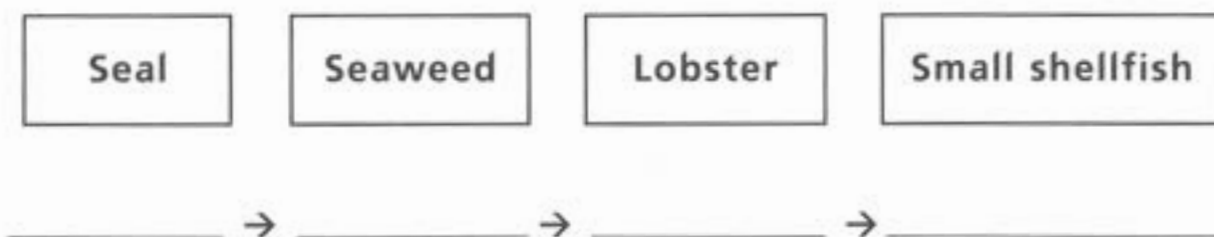


The seaweed (a form of algae) is a producer. The crab (a crustacean) is a **first-order** consumer because it eats the seaweed. The seabird is a **second-order** consumer because it eats crabs.

- 3 The seaweed is a marine organism. What is meant by *marine*?
- 4 If this food chain were extended to include a third-order consumer, what would this third-order consumer be likely to eat?

Food chains show how energy passes from producers to consumers as members of the chain are eaten by other members further up the chain. As each food is digested, energy is released.

- 5 Does the food chain above show all the organisms involved? Explain your answer.
- 6 What is meant by the arrows shown in the food chain?
- 7 The organisms in a food chain below are not in the right order. Write them in the correct order.



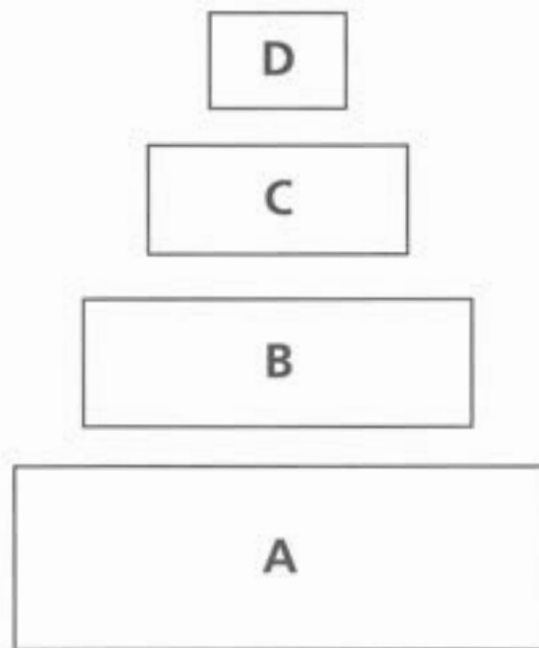
- 8 Give an example of a *crustacean*.
- 9 Outline the difference between a *heterotroph* and an *autotroph*.

Food pyramids

As you move further up a food chain, the consumers become larger in size but fewer in number. The amount of *biomass* (biological mass or mass of living things) decreases as you go up the food chain.

In the model of a food pyramid to the right, **A** represents leaves and vegetation, **B** represents insects, **C** represents small birds and **D** represents raptors (birds of prey, eg eagles and hawks.)

Green plants make their own food from solar energy, carbon dioxide and water. When an insect eats a leaf, it uses up some of the energy in moving, growing etc. This means that when the insect is eaten by a small bird not all of the energy is passed to the bird from the leaf because some of it was incorporated into the insect. Therefore the bird needs to eat many insects to get enough food to support its needs. Consequently there will be many more leaves than insects and many more insects than small birds. There will also be fewer birds of prey than small birds.



10 Write out the food chain described in the food pyramid above.

11 Name the producer.

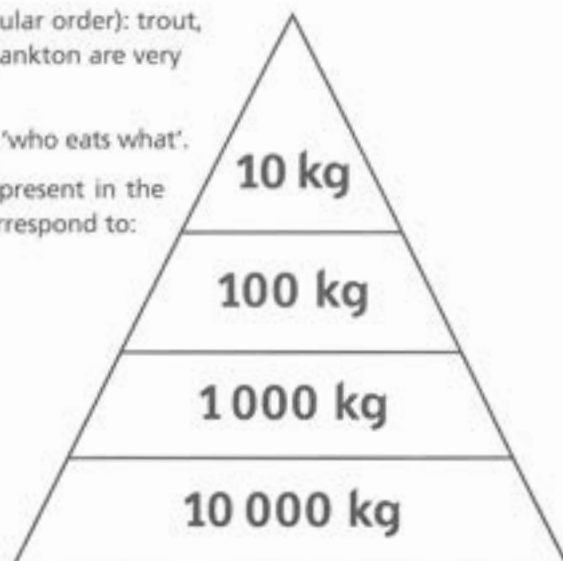
12 Name the third-order consumer.

13 What is a *raptor*?

The following organisms are present in a small lake (in no particular order): trout, insect larvae, fingerlings (small fish) and phytoplankton. Phytoplankton are very small plants that have chlorophyll and can photosynthesise.

14 Write out a food chain for the organisms in the lake, showing 'who eats what'.

The diagram on the right shows the masses of each organism present in the lake. Give the names of the organisms that are most likely to correspond to:



15 10 000 kg

16 1 000 kg

17 100 kg

18 10 kg

A person came along and started to fish in the lake. Eventually she caught all of the trout present.

19 Forecast what might happen to the number of fingerlings. Explain your answer.

20 What are *phytoplankton*?

21 What is *biomass*?

22 There are some organisations in Australia that release millions of young fish into water systems each year. Give a reason why they might do this.

Flows of energy through ecosystems

The word *trophic* refers to nutrition and *trophic levels* are the different levels through which nutrition flows by way of the food pyramid or food chain. Organic matter, called *biomass* (*bios* living, *mass* a quantity of matter), travels through the food chain starting at the level of primary producer and peaking at the top carnivore.

1 Explain what is meant by a *food chain*.

The amount of available biomass (and energy that comes with it) is reduced at each level above the primary producer; in other words, as it travels through the food chain. This happens because some of the energy is incorporated into the organism that eats it. For this reason, a food pyramid tends to reach a maximum of five levels, usually fewer.

At the most basic level we find the *primary producer*. Primary producers can create their own food by converting the sun's energy via a process called *photosynthesis*. Primary producers are found on land (terrestrial) and in water (aquatic).

2 Give an example of a *terrestrial primary producer*.

3 Give an example of an *aquatic primary producer*.

The trophic levels above the primary producer level are collectively called *consumers* because they obtain nutrients and energy by consuming other organisms. However, consumers can be classified in several ways. At the second level of the food pyramid we find the *herbivores* (plant-eaters) also called grazers. These include all species that do not eat meat, including plant-eating insects. The levels above the herbivores are the *carnivores* (meat-eaters) and there can be several ranks.

4 Name two native Australian herbivores.

The term *carnivore* usually refers to the first meat-eating level, but since carnivores can also be eaten, a *secondary carnivore* and a *tertiary carnivore* (sometimes called *super carnivore*) are included in some classifications. For example, a snake eats other animals but owls can also eat this meat-eater. In the case of two fish-eating animals, the shark and the seal (excluding leopard seal), both are considered carnivores. However, since the shark would just as easily eat the seal as eat a fish, but the seal would not eat a shark, the shark would be the secondary carnivore, placed above the seal on the pyramid. There are not many species of tertiary carnivore. They include humans, killer whales and leopard seals. Tertiary carnivores are seldom hunted as prey.

The Great skua (*Stercorarius skua*) and the Australian dingo (*Canis sp.*) provide good examples of the complexities of the food web and how an organism can be classified in more than one position. The Great skua is a large predatory bird, generally found in the colder parts of the Northern Hemisphere (ie Iceland, Scotland), Antarctic and southern South America. The Australian dingo is a wild dog that prefers to feed as a carnivore, however, its success lies in its flexibility when it comes to diet. Dingoes have been seen eating fish, seaweed and other plant matter when meat is unavailable.

5 How would you classify the skua when it hunts fish, like other fish-eating birds, but also hunts fish-eating birds? Explain your answer.

6 How would you classify a dingo on the food pyramid when it eats meat and when it eats plant matter? Explain your answer.

7 Give the general name for an animal that eats both plant and animal matter.

8 Would you consider it accurate to classify the dingo as the name given in your answer to question 7. Explain your answer.

Components of an environment – biotic and abiotic

The biotic components of an environment are the organisms present.

1 What are the abiotic components of an environment?

For scientific purposes, the living world has been divided into groups called *biomes*, meaning 'zone of life'. Biomes are large, naturally occurring communities of plants and animals forming recognisable groups occupying a large territory. The two major world biomes are aquatic and terrestrial. They can be divided into more specific categories based on the species present.

Aquatic (*aqua*, water): These are the 'wet' biomes and consist of freshwater (eg ponds, rivers) and marine (eg sea/ocean, estuary). The composition of the water, which differs between each, affects the kinds of species that will be found.

2 What is the main difference between freshwater and marine conditions?

Desert (*desertare*, to forsake): Roughly 20% of the Earth is desert. A biome that commonly receives less than 50 cm of precipitation per year is classified as a desert. This lack of water means that many types of vegetation cannot survive and deserts are recognisable by the vast stretches of sand on which little or no vegetation is present. Any vegetation present has evolved to adapt to the arid conditions.

Forest (*forestis*, wood): Forests are biomes that are densely covered by various tree species (depending on their geographical location). Smaller trees and shrubs are usually found beneath the tree canopy. Forests cover about 30% of the Earth and represent 70% of the carbon found in living organisms.

Tundra (*tanturia*, treeless plain): Commonly found in cold northern regions of North America, parts of Asia and the Arctic districts of Europe, tundra are large, treeless areas. At ground level, the soil is cold and hard. As expected, these are the most frigid of the biomes. *Arctic tundra* is found in the Northern Hemisphere (eg North Pole) and may be flat or slightly rolling – but not mountainous, while *alpine tundra* is that which has an alpine, or mountainous, landscape. The combination of high altitudes and cold temperatures makes it impossible for trees to grow on alpine tundra.

Grassland: As the name suggests, grasslands are large areas covered by grasses.

Savanna: Savannas are often mistaken for grasslands but they can be distinguished by the presence of an occasional tree or shrub. More than half of Africa's landscape is savanna. The average rainfall is 50–130 cm per year, most of which comes in a fairly short space of time followed by long periods of hot, dry conditions.

3 The North Pole and Antarctica have often been compared to desert. Suggest a reason why.

4 Give a definition for *arid*.

5 Give a definition for *canopy*.

6 Suggest a meaning for *graminivorous* then check with your dictionary to see if you were correct.

Give definitions for the following biomes, including naming an area where each can be found:

7 mangrove

8 reef

9 rainforest

Research

10 Investigate and describe what *spinifex* and *tussock* are.

11 Another type of biome is the *sclerophyll forest*. Briefly describe the differences between wet sclerophyll and dry sclerophyll (do not simply state that one is wet, the other dry – describe rainfall and vegetation).

Bioaccumulation and biomagnification

In the same way that nutrients and other compounds can be cycled through an ecosystem, so can harmful toxins and pollutants. The terms *bioaccumulation* and *biomagnification* are often interchanged but each has a specific meaning. Both words refer to the effects of pollutants (or toxins) entering the food chain. The difference lies in their effects once they are present.

Bioaccumulation describes how substances enter the food chain at the lowest trophic level, in other words, from the grazer upward to the final trophic level, the top carnivore. Bioaccumulation describes the accumulation of an undesirable substance *within the tissues of an organism or a population of organisms*.

Biomagnification (sometimes called *bioconcentration*) describes the magnified effects, or increasing levels of concentration, of these compounds as they travel along the food chain (trophic levels). Consider the simplified example below used to illustrate how the amount of toxin can become biomagnified within an ecosystem:

A certain water bug is considered a pest in Lake Perfect. The water is treated with a toxic chemical compound that kills most of the bugs. One molecule of poison is sufficient to kill each bug, therefore each dead bug contains one molecule of the toxin.

Ten 'mini-fish' come along and each fish eats 50 poisoned bugs. The tissues of each mini-fish then contain 50 molecules of toxin. A large 'killer fish' then eats 10 of the mini-fish (each of which contains 50 molecules of toxin). The tissues of the killer fish then contain 500 molecules of toxin.

A predatory bird eats 5 of the killer fish. Its tissues will contain 2500 molecules of toxin. A land carnivore kills and eats 3 of the predatory birds (each of which has eaten 5 killer fish). The tissues of the land predator will contain 7500 molecules of toxin.

Thus, the compound used to kill the water bugs was biomagnified from one molecule per individual, at the lowest trophic level, to 7500 per individual at the top level as it travelled up the food chain. In some instances, substances can bioaccumulate at one level only but the organism's metabolism is able to break it down. Because of this, it may not become biomagnified throughout the whole food chain.

- 1 Show in figures how, in the example above, one molecule of toxin magnified to 7500.

Give meanings, in biological terms, for these words:

- 2 uptake
- 3 ingest
- 4 storage
- 5 eliminate
- 6 equilibrium
- 7 Describe the difference between *bioaccumulation* and *biomagnification*.
- 8 Suggest other ways in which scientists might look for evidence of bioaccumulation besides testing animal tissue.
- 9 A certain branch of science is called *ecotoxicology*. Suggest what ecotoxicologists do.

Identify the following terms mentioned on this sheet.

- 10 A hierarchical layer of a food web.
- 11 Flesh-eating mammal.
- 12 A substance that provides energy and vital substances to an organism.
- 13 Combination of organisms and the area in which they live.

Heavy metal pollution: the Minimata case

Mercury (chemical symbol Hg) is a 'heavy metal', one of a set of metal elements that are toxic. Mercury has been used for thermometers, barometers, electrical switches, lighting, batteries and in the manufacture of sodium hydroxide. The metal itself is moderately toxic, but becomes severely toxic when it forms compounds with carbon.

Early last century, the people of the Japanese fishing village of Kumamoto, about 600 km from Tokyo, hoped to expand their quiet little fishing village to a larger, more prosperous town. Consequently they welcomed the arrival of the chemical producing company, Chisso Corporation. The Chisso plant produced, among other forms of waste, mercury compounds that were freely dumped into Minimata Bay. The villagers depended on this bay for fish. Almost immediately the deadly effects the chemicals had on the fish became evident. The company recognised the damage and paid the villagers for permission to ruin their fishing industry. The dumping of mercury compounds continued from the late 1920s up until 1968. Minimata Bay flowed out to the Shiranui Sea (an inland sea), poisoning the aquatic species along its path.

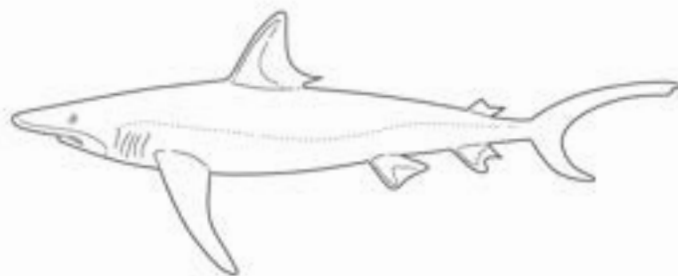
By the 1950s rumours began to spread that the village was haunted by a spirit that took over the bodies of Kumamoto residents. Many cats, and some dogs, began to do 'weird dances' on the streets before dropping dead. Children were also behaving strangely and soon adults followed. People began to develop speech and vision problems, others began screaming wildly, muscle spasms led to paralysis and suicides occurred. The bizarre condition became known as '*Minimata Disease*'.

People then suspected some form of infectious disease was attacking the central nervous system. The corporation's doctor, Dr Hajim Hosokawa, discovered the connection between a fish diet and the condition. This led him to analyse the fish and identify the heavy mercury contamination.

- 1 Chisso Corporation offered compensation pay-offs in exchange for affected people signing waivers that surrendered their rights to court action. Give reasons why the company would be willing to pay the villagers to destroy their fishing industry.
- 2 Suggest why children and animals were the first to be affected by the toxin.
- 3 Would the effects on the villagers have been the same if the Shiranui Sea not been an inland sea? Explain your answer.
- 4 Give two reasons why local residents might welcome the arrival of a chemical company.
- 5 Name two uses for mercury.

A television current affairs program ran a 'special' on the effects of mercury on pregnant women. One piece of advice from a researcher was to advise pregnant women not to eat fish such as swordfish or sharks.

- 6 Why would the researcher specifically mention these two types of fish?



Research

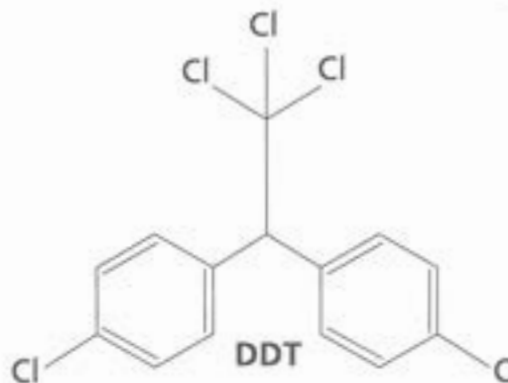
- 7 The Mad Hatter in '*Alice's Adventures in Wonderland*' (Lewis Carroll, 1865) shows classic signs of mercury poisoning. Find out how the phrase '*Mad as a hatter*' originated.

DDT: the Peregrine Falcon case

16

A chlorinated hydrocarbon compound known as *dichloro diphenyl trichloroethane*, or DDT, provided one of the earliest examples of the effects of biomagnification. DDT is a colourless, odourless and very effective insecticide that first came into use in the 1940s.

DDT degrades slowly, with a half-life of 15 years, so it is considered a non-degradable substance. It is not water-soluble but, being fat-soluble, is taken up by fatty tissue within the body. DDT illustrated the sub-lethal effects of biomagnification. *Sub-lethal effects* are those that *indirectly* cause death.



- 1 Define *insecticide*.
- 2 What is meant by *half-life*?

The peregrine falcon is one of a number of predatory birds belonging to the raptor family. During the decade from the mid-1950s to mid-1960s, the numbers of surviving falcon chicks dropped dramatically. Tests revealed that the parent falcons contained high levels of DDT yet they appeared able to cope with it – so what was the problem? Upon inspecting the nests, it was observed that the shells of the falcon eggs were unnaturally thin. During normal incubation, the parents could not avoid breaking the paper-thin shells. It became obvious that DDT levels were directly affecting the hen's ability to produce a thick enough shell.

DDT did not directly kill the chicks, but indirectly killed them by preventing a strong enough eggshell in which to develop. Similar effects have been observed on Bald Eagles and other predatory birds. The prey does not contain sufficient levels of DDT to harm them but the birds accumulate the toxin with each new feed. Fortunately, the establishment of captive breeding programs has enabled the peregrine falcon to make a comeback and the use of DDT has since been banned in most developed countries.

- 3 Suggest how the peregrine falcons accumulated so much DDT in their tissues.

DDT is a 'broad spectrum' pesticide, meaning that it affects a broad range of pest species and not just the target species. There have been many documented cases where the pesticide was sprayed on crops to destroy insects that ate the crop, for example locusts. Birds and lizards swarmed on the easy meals that lay dying on the ground. Soon after the birds and lizards began to die from DDT poisoning, as well as those that preyed on them, such as cats or larger birds and reptiles – another example of biomagnification at work.

At the same time other insect species moved in to eat the crops. Normally these species would be kept under control by wasps, birds and other predators but, since DDT also killed those *non-target species*, the new arrivals were free to replace the original pests. Compound this with a few of the surviving original pests developing resistance, and the whole cycle repeats itself, minus some helpful, naturally occurring predators.

- 4 What is meant by a '*broad spectrum*' pesticide?
- 5 Explain why the fact that DDT is fat-soluble makes it so dangerous to vertebrates.

Integrated pest management: DDT and pest control

To *integrate* means to incorporate a number of factors, or to form a whole unit out of different component parts. *Integrated Pest Management (IPM)* incorporates various techniques to produce a management strategy that is the safest, most efficient possible for a given situation. After the effects of DDT became apparent, IPM strategies were recognised as the sensible alternative.

Research

- 6 Find out techniques that are used in IPM.
- 7 Read 'Silent Spring' (Rachel Carson, 1962) and comment on its relevance to this topic.

Parasitism, mutualism and commensalism

Relationships between organisms take several forms. *Symbiosis* is the term used to describe how different plant or animal species live together. The effects of the relationship are then classified as *parasitism*, *mutualism* or *commensalism*.

Parasitism

The word *parasite* comes from an old word meaning 'one who eats at another's table'. A parasite is an organism that obtains its nutrients by living in or on another organism. The parasite obtains nutrients at the expense of the host organism it associates with. Thus, *parasitism* involves one organism depriving another of its nutrition. In this case, the parasite benefits and the host suffers. An example of a parasitic relationship is the tapeworm. Tapeworms can infect humans and other animals, living in the alimentary canal of the host. The tapeworm takes much of the nutrients its host ingests, leaving the host undernourished as a result (and in extreme cases, dead).

Mutualism

As the name suggests, *mutualism* involves a relationship that is of mutual benefit to both organisms. A well-known case of mutualism involves various species of birds living on or near larger animals, such as horses, cows, rhinos and giraffes. The birds eat the tiny bugs that would otherwise irritate or damage the skin of the larger animal. In exchange for providing this easy meal, the larger animal benefits by having its body kept free of pests. Similar examples occur in the oceans with smaller fish performing the same task on larger species. For example, a remora is a fish that has a disk on its head, which it uses like a sucker to cling onto other fishes, especially sharks. Remoras feed on food scraps left over by their hosts or on parasites of the host. They can also swim alone and have been seen clinging to the hulls of ships.

Commensalism

Commensalism describes a relationship in which one species benefits from another by way of obtaining food and/or shelter, but the other species receives no benefit at all. Commensalism differs from parasitism because the host organism is not harmed in any way.

- 1 Would you describe the relationship between a shark and a remora as *parasitism*, *mutualism* or *commensalism*. Explain your answer.

The digestive tracts of ruminants (ie cows, horses) contain cellulose-digesting bacteria.

- 2 Name the type of relationship the two species have.
- 3 How do the ruminants benefit from the relationship?
- 4 How do bacteria benefit from the relationship?

Lampreys and hagfish (*Class Agnatha*) are fishes with sucker-like mouths that attach to the host. Their rasping tongues are used to scrape away at the flesh of the fish, enabling them to feed on the tissue of their host.

- 5 How would you classify lampreys and hagfish?
Explain your answer.

The common European cuckoo (*Cuculus canorus*) is a bird that lays its eggs in the nests of other birds which currently have eggs. The cuckoo's offspring are brought up by the original inhabitants of the nest – often causing their offspring to suffer.

- 6 How would you classify the cuckoo? Explain your answer.

Identify the terms defined by the following statements:

- 7 In *antagonistic symbiosis*, this is the organism that suffers by satisfying the needs of another.
- 8 Hoofed mammals that chew the cud and have a complex four-chambered stomach.



The environment a species inhabits will affect how it adapts and evolves. Dispersion, or 'spreading-out' activity, plays an important role in many behavioural adaptations. Dispersion (*disperse* to part) can involve animals or their food supply. For example, an animal that must travel long distances at regular intervals in order to eat will need, among other characteristics, the strength and stamina to withstand the trips. Natural selection will favour individuals who possess these features.

Other ecological adaptations are associated with daily life. Flexibility is important. Some important factors to consider include:

- Is it more advantageous for the species to live in groups or alone?
 - If the species lives in groups, is it better to work together as a co-operating community (eg ants, bees), or to just live as individuals who happen to share a habitat? This point applies to territory as well. Should the animals defend their whole territory as a group or just defend their own small portion of it from others (whether they are members of the community or not)?
 - Should parental care be long-term or non-existent? For long-term care, would it be more beneficial for one or both parents to care for their own offspring or, as in the case of lionesses, the whole group protects the young, with several females feeding cubs whenever possible.
- 1 The ability to adapt one's behaviour helps to maximise an organism's fitness. Discuss this statement.
 - 2 Describe the difference between *adapted* and *inherited*.
 - 3 Suggest ways in which available foods can lead to behavioural adaptations.
 - 4 Suggest ways in which available foods can lead to evolution of a trait.

The tiger salamander (*Ambystoma tigrinum*) has a cannibalistic form, or *morph*. When its aquatic environment becomes overcrowded and food shortages occur, certain of the larger individuals turn cannibalistic, eating smaller members of their own species.

- 5 Suggest how these morphs might maximise the survival of the species within the pond.
- 6 Discuss how the different types of social organisation might affect how a species adapts.
- 7 Discuss two advantages and two disadvantages of the different types of parental care.

Identify the terms defined by the following statements:

- 8 *Organism that eats its own kind.*
- 9 *Place where an organism usually lives.*
- 10 *Process of gradual change to a better state.*

Jean Lamarck

Many people have noticed that animals change over time and have come up with explanations for these changes. Jean Lamarck (1744–1829) proposed that if an animal changed during its lifetime these alterations were passed on to the offspring. The most quoted example of his theory is that of how giraffes gained their long necks.

Giraffes have long necks especially suited to grazing on tall trees. According to the Lamarck theory, giraffes developed long necks by a gradual process. In times of plenty, all giraffes ate well. When drought came and forage was in short supply, giraffes stretched to eat any food that was available and gained long necks through the stretching. Those giraffes with long necks survived and passed on their characteristics to the next generation, who continued to stretch for food and get longer necks. Over many generations giraffes developed long necks.

- 11 Most dogs have tails. Some dog breeders dock (cut off tails) young dogs because the particular breed might be more fashionable without a tail. Explain why no matter how many tails of young dogs are removed, the offspring will still have tails.

Fossils reviewed

Palaeontology is the science of fossils (*paleo* ancient). A *fossil* can be defined as any trace of a plant or animal that has been preserved in the Earth's crust, most commonly in rock or sand. Fossils can be formed as imprints, skeletal remains or by an organism being petrified. Often only the hard parts of organisms (living things), for example bones, shell or bark tend to be preserved.

A fossil can be made from a porous (contains lots of small holes) object such as a bone. Under the right conditions water containing minerals can fill the porous parts. Over a long period of time this can make the object resemble stone and preserves it. In a similar way, a piece of wood can have its molecules replaced very slowly by minerals, making it petrified.



1 *Petrified* literally means 'turned to stone'. Give another meaning of *petrified*.

In rare cases, such as examples of the Woolly Mammoth, entire bodies (including stomach contents!) have been found preserved in Europe.

2 Which modern (but less hairy) mammal does the Woolly Mammoth resemble?

3 Suggest why *palaeontology* and *geology* are closely linked.

4 Suggest why, in places like Siberia, entire bodies have been preserved.

Fossils form when materials from the Earth's crust (sediment) cover the remains of an organism, either under water or on land. As greater amounts of deposit accumulate, the remains are compressed by the rock. Imprints (perhaps a footprint in soft soil) can also be filled in and covered up over time.

Various scientific tests can be performed to determine the ages of fossils, thus enabling us to develop theories as to how life on Earth evolved. Not only does finding a certain species in a given area suggest what lived there (and when), but it can also tell us about environmental conditions. For example, fossilised shells have been found in deserts suggesting that water once covered the land.

5 Many insect remains have been preserved in amber, a hardened, golden-brown tree resin. Suggest why these trapped insects did not decay.

Sometimes only part of an organism is fossilised when it dies.

6 What parts of plants are most likely to be found fossilised? Explain your answer.

Imprints, such as tracks and footprints, can form fossils. One example might be a dinosaur walking on a muddy riverbank leaving its footprints. The mud dries out and becomes covered in sand and other sediments that eventually form hard rock. Millions of years later the original footprints become uncovered. The size and distance between these footprints can give information about the dinosaur.

7 Explain how this information might help a palaeontologist work out the physical dimensions (weight, height etc) of a dinosaur.

Research

Fossils are classified into four groups. Investigate the following types then write short descriptions of how they are formed.

8 Casts and moulds 9 Indexed remains 10 Replaced remains 11 Impressions

Remember! If you find or collect fossils, be sure to record the location and date the fossil was found. Your fossil might one day help scientists to answer important questions about life in the past.

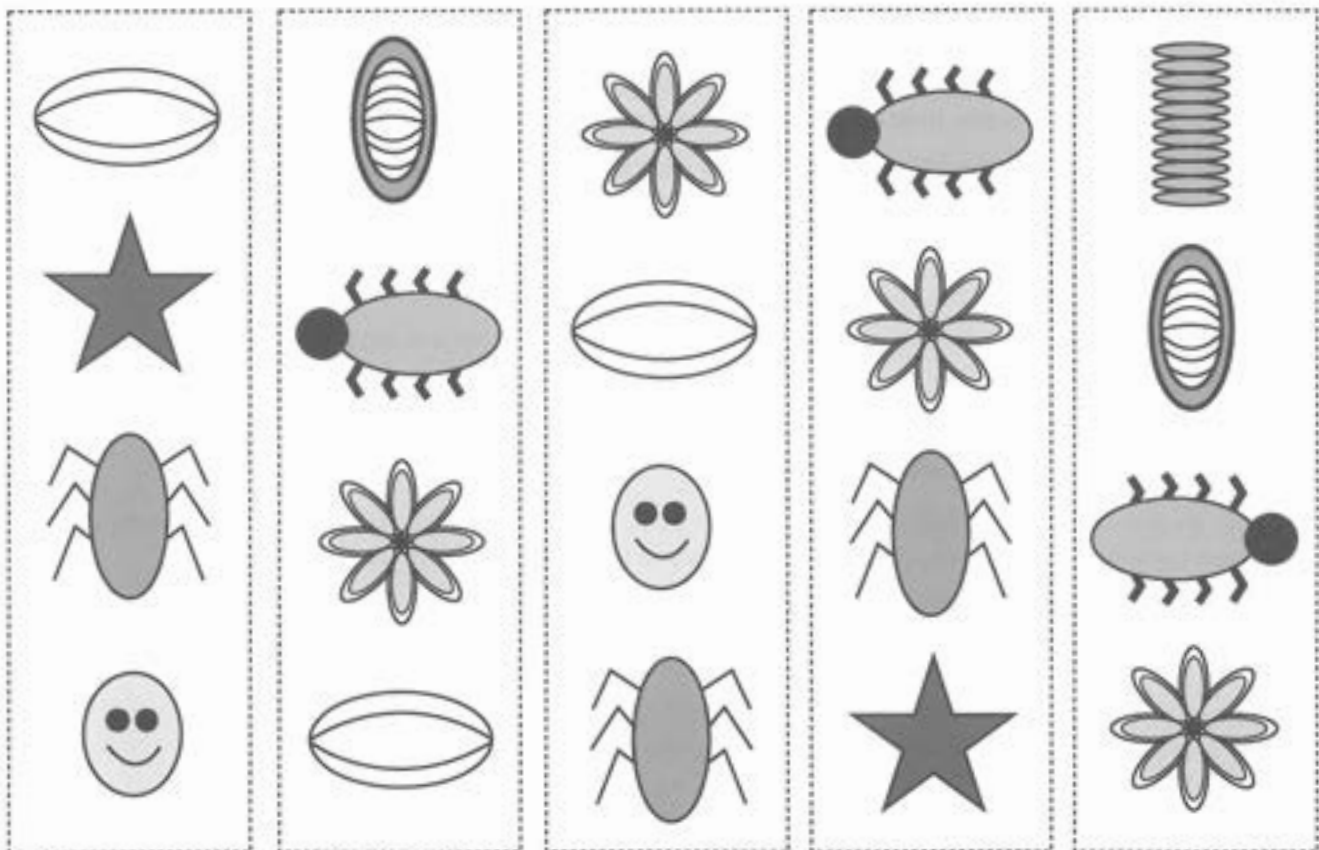
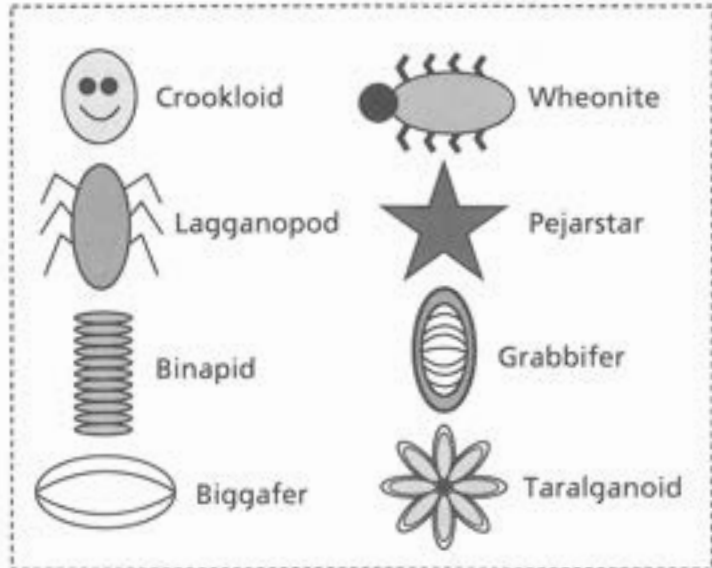
Strata and fossils

A group of geologists went digging for fossils in five locations (A, B, C, D and E) all within 2 kilometres of each other. The key to the right identifies the various types of fossils found.

- 1 Study the key and then cut it out and stick it in your book.

The fossils seemed to belong to strata (layers) in a pattern. The five rock strata samples that the geologists found are shown below.

- 2 Cut out and match up the strata. You should find that there is one layer that doesn't seem to agree with the rest. Put this to one side.
- 3 Stick the *four* strata that do line up into your book, with fossils matched up.
- 4 Label each fossil using the key.
- 5 How many different types of fossil are there?
- 6 How many fossil layers were there in total?
- 7 Which is the oldest fossil in the strata? Give a reason for your answer.
- 8 Which is the youngest fossil in the strata? Explain why.
- 9 How do you think that the 'odd man out' occurred?



Did you know? The earliest evidence for life on Earth was found at the Pilbara Region (West Australia).

Vertebrate evolution

Before you attempt this section, remind yourself of the following classification terms.

Class, Family, Genus, Kingdom, Phylum, Order, Species

1 Rewrite them in order, from the least specific to the most specific.

Many species in the phylum *Chordata* are considered the most advanced of animals. It was long believed that the chordates evolved from the *Echinoderms*, meaning 'spiny skinned animal' (eg starfish). However, most scientists now believe the two groups co-evolved from a different, common ancestor. Since most of the early chordates were soft-bodied and believed to have arisen about 700 million years ago (during the Pre-Cambrian era), it is near impossible to find fossil evidence to confirm what embryonic development patterns suggest about their relationship.



2 What does co-evolve mean?

A distinguishing feature that separates the first vertebrates from the chordates is the presence of a *notochord* at some point during development. The notochord is a collagen-based rod that strengthens the dorsal (back) part of an animal and develops into a vertebral column (backbone). Collagen is a fibrous protein that can make up connective tissue. The notochord commonly extends from the tail to the head and, as the animal develops, it is replaced with the vertebral column. The original chordates were believed to be slender, soft-bodied, filter-feeding animals. The earliest chordates were the ancestors of the vertebrates. The first vertebrate form was the *Agnathans*, or jawless fishes. The hagfish and lampreys are examples of jawless fishes but, interestingly, the two groups did not evolve from the same ancestor.

3 Lampreys exist on the blood of other fishes. Which of the following could not be used to define lampreys according to the description of *Agnathans*?

A Parasites

B Marine

C Carnivore

D Invertebrate

The phylum *Chordata* evolved into three branches:

- *Urochordata*: These are often found in colonies and are commonly sessile (not free to move about). Examples include tunicates (these are covered in a protective 'tunic'), sea squirts, cunjevoi, sea cucumber.
- *Cephalochordata*: This 'middle' group is in many ways similar to the vertebrates in having evolved a circulatory system. However, this species do not have a true heart, for example amphioxus.
- *Vertebrata*: This most advanced group possesses a heart, brain, circulatory system and vertebral column, for example humans.

Radial symmetry is when there are similar parts regularly arranged around a central point. Vertebrates are bilaterally symmetrical.

4 Give a definition for *bilateral symmetry*.

5 Compare *bilateral symmetry* with *radial symmetry*.

6 To which phylum do we belong? We include sea urchins, brittle stars and sea cucumbers. We have an internal skeleton, often have five parts, are marine animals and sometimes have spikes.

7 Give a reason why there are few fossils of early chordates.

Research

8 Jean Lamarck (1744–1829) came up with a theory of evolution 50 years before Charles Darwin. Give a brief summary comparing Lamarck's theory of evolution with Darwin's.

Researchers are divided as to exactly when the primates first appeared on Earth. Some believe they evolved from an insectivorous ancestry during the late Cretaceous (135–70 million years ago) period. Others date the first primates at the Palaeocene (65–54 million years ago) period. Either way, the earliest primates were here a long time ago.

- 1 Define *insectivore*.

Primate groups

There are two divisions of primates. They are the *prosimians* (*Prosimii* – *pro* before, *simian* apelike) and the *anthropoids* (*Anthropoidea*, resembling man). There are five families of the older *prosimii*, or 'Old World' primates, most of which are found in Africa, the Philippines and Madagascar. Examples of *prosimians* are the lemur, tarsier (related to the lemur), loris, and the tree shrew.

Primates are animals, which belong to the mammalian order primates.

The *anthropoids* include the monkeys, apes and humans. Their faces are generally much flatter than the *prosimians* and they have two premolar teeth. *Anthropoids* also have areas on their faces and/or bodies that are not covered by hair. They can sit in an upright position – or at least on their haunches (the area consisting of the thighs and buttocks). *Anthropoids* use their opposable thumbs and grasping hands to hold objects and aid in climbing. Depending on the species, they can walk either completely, or for short periods, using bipedal (two-legged) motion. Apes lack a tail.

The 'New World' *anthropoids* include the monkeys and marmosets. The monkeys and marmosets are mostly arboreal and lack opposable thumbs, tending to use their prehensile tail to grasp (*prehensile* to seize). New World *anthropoids* walk using their arms and legs (quadrupedal). They have three premolars, as opposed to the pair found in *anthropoids*.

Important note: We often describe primates as 'Old World' or 'New World'. The terms 'old' and 'new' refer to the landmasses on which the species are found. It does not refer to the evolution of the species. The Old World is the landmasses we have known about longer, including Africa and Asia. The New World, discovered much later, is the Americas.

Primate evolution

Primates have evolved with time. As the arboreal (tree-dwelling) *prosimians* evolved, their eyes moved from the side to the front of the head, giving them binocular vision. This means that they can judge distance and depth more accurately. They also developed hands with separate fingers and opposing thumbs that could grasp branches. Some *prosimians*, such as the ring-tailed lemur, found in Madagascar, travel on all fours, with knuckles of the hands touching the ground. Other lemurs are arboreal. The big-eyed loris and the tarsier of Asia have large eyes that are set in a fixed position. These animals must turn their heads (by up to 180°) to look to the side.

- 2 Name two examples of primates that have a terrestrial (for the most part at least) rather than arboreal lifestyle.
- 3 State an advantage of binocular vision.
- 4 State the meaning of *arboreal*.
- 5 Explain why opposable thumbs are so useful.
- 6 What is meant by *bipedal motion*?

Research

- 7 Scientists do not always agree as to whether the insectivorous tree shrew should be classified as a primate. Research the features of the tree shrew. Do you agree or disagree that it should be classified as a primate? Give reasons why.



Monkeys

Monkeys first appeared in Africa about 30 million years ago. They were a more advanced type of primate, and were more successful in competing for living space and food than lemurs, and began to take over from the lemur. (Since Madagascar had become an island by then, monkeys did not reach there. Hence the lemurs are more numerous on Madagascar.)

During their evolution primates have become larger. New World monkeys tend to be smaller than Old World monkeys. Many have a prehensile tail that acts as a useful fifth limb. In the Old World some monkeys, such as the baboon, have left the trees and become almost completely terrestrial, ie stay on the ground. The slower moving orang-utans (*'the old man of the forest'*) tend to stay in trees. The exceptions are older and larger males who travel on the ground.

8 Explain how a prehensile tail can be useful for a monkey.

The gibbons of Southeast Asia swing and leap rapidly through trees using their long arms. When on the ground they can walk upright, using their arms for balance.

Africa is home to the chimpanzee and the gorilla. The gorilla is the largest primate with an average weight of 270 kg. Both chimpanzees and gorillas climb trees, especially to nest at night, but are more terrestrial than Asian apes. All of the *great apes* (orang-utans, chimpanzees and gorillas) are able to walk upright but more often travel on all fours.

9 Explain why some primates who generally live on the ground during the day, might nest in trees at night.

All monkeys have eyes at the front of the head, giving them binocular vision. Having excellent colour vision, they can see well through vegetation. The characteristics gained from living in trees have been a great help to those monkeys now living on the ground. These skills include good eyesight, an ability to learn quickly from experience and manipulative hands.

10 What is meant by *manipulative hands*?

As primates evolved their bodies grew in size and so did their brains. Approximately 3 million years ago the first *hominids* (apes resembling humans) appeared. The shapes of the hips and pelvis suggest that they walked upright on two legs.

Primate social patterns

Primates exhibit a wide range of social patterns. They can communicate through sounds, facial expressions, making signals and posturing. Some, like the gibbons, live in family groups. Howler monkeys and chimpanzees form large, loosely organised groups. A typical orang-utan group has one or two females with young, with males joining them only for mating. Gorillas live in family units. Different species look after their offspring for different lengths of time. Tree shrews need to care for their young for approximately 3 months, great apes for about 7 years and humans need even longer periods of time.

11 Explain what is meant by *posturing*.

12 Discuss the statement: *'The more evolved the organism, the longer the developmental time of its offspring'*.

The *Apes*, including the gibbon, chimpanzee, orang-utan and gorilla (*Pongidae*), have broad chests, arms longer than their legs and lack tails. They are larger and more intelligent than other primates, can use simple tools and have even been taught to communicate using sign language*. They are mainly vegetarian, with the exception of the gorilla, who will occasionally eat small animals.

Research

13 *Use the Internet to research the fascinating story of *'Koko the Gorilla'*.

14 The sloth is a South American mammal that has many features in common with some of the primates we have mentioned. Find out about the sloth and decide whether or not it is a primate.

Primates revision questions

- 1 Name the two major divisions of primates. _____
- 2 Define *quadrupedal*. _____
- 3 If the 'Old World' is Africa and Asia, what is the 'New World'? _____
- 4 Identify three characteristics that distinguish apes from other primates. _____
- 5 Which primate looks after its offspring for the longest time? _____
- 6 Name the sense that is most valuable to a monkey. _____
- 7 Which is the largest of the primates? _____
- 8 Name three primates that can walk upright. _____
- 9 When did the first hominids appear? _____
- 10 Give *three* ways in which primates can communicate with each other. _____

Decide whether the following statements are true or false.

- 11 All primates are mammals. _____
- 12 Prosimians tend to exhibit solitary behaviour. _____
- 13 Anthropoids tend to form groups. _____
- 14 There are five families of prosimii. _____
- 15 Humans are anthropoids. _____
- 16 Anthropoids have two premolar teeth. _____
- 17 Anthropoids have opposable thumbs. _____
- 18 Prosimii lack tails. _____
- 19 Monkeys and marmosets are New World monkeys. _____
- 20 The prosimian's eyes were once on either side of the head. _____
- 21 Anthropoids can spend some of the time using bipedal motion. _____
- 22 The monkey is one of the more advanced primates. _____
- 23 New World monkeys tend to be larger than Old World. _____
- 24 Humans are the largest of the primates. _____
- 25 Gorillas live in family units. _____
- 26 Gorillas live in nests. _____
- 27 As primates have evolved, they have become smaller. _____
- 28 Primates first appeared on Earth a few hundred thousand years ago. _____
- 29 Most apes tend to eat mainly vegetation. _____
- 30 The word *anthropoid* means 'having two feet'. _____

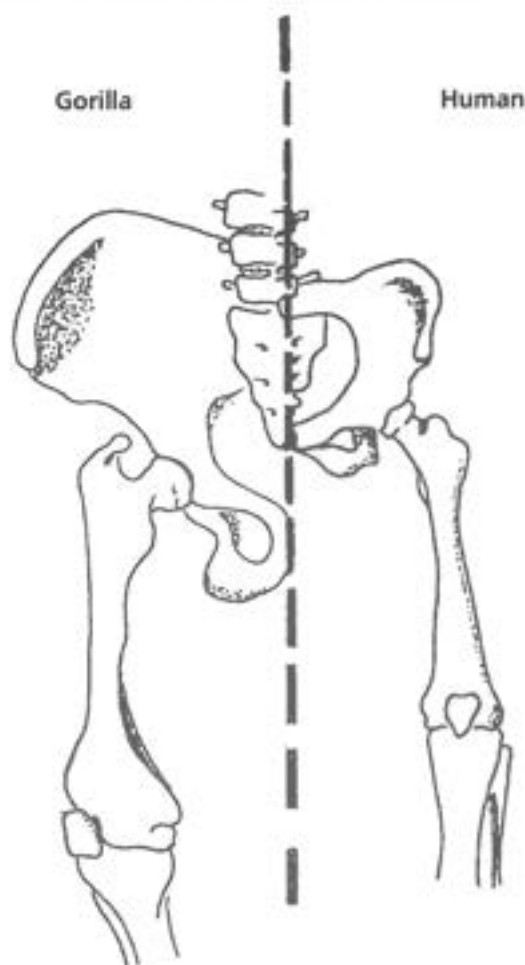
Homo sapiens

A species is made up of related organisms capable of interbreeding. They have a two-part name: the genus (first letter capitalised), then a Latin species name (no capitals). Scientists classify all humans currently living on Earth as members of a single species, *Homo sapiens*. The list below shows how we can classify *Homo sapiens* based on anatomical features:

- kingdom **Animalia** – an animal
- phylum **Chordata** – with a backbone
- subphylum **Vertebrata** – segmented spinal column
- class **Mammalia** – suckles its young
- subclass **Eutheria** – offspring develop with a placenta
- order **Primate** – hands/feet with five digits and two mammary glands on the chest
- suborder **Anthropoidea** – stereoscopic vision and a proportionately large brain
- family **Hominidae**

The primates that most closely resemble humans are chimpanzees, gorillas and orang-utans. Skeletons of *Homo sapiens* differ from these because of a very early adaptation to standing upright (erect posture) and walking on two feet (bipedalism). Humans have an S-shaped spinal column, putting the centre of gravity of the human body directly above the feet. This, and the modified pelvic region, makes walking upright easier.

Other anthropoids can walk upright to a limited extent, however, they all have spines which are straight or arched. They also have prehensile (grasping) feet and use their front limbs (hands) to lean on when walking. Because humans walk completely upright, hands have become much more developed, for example the thumb is fully opposable to the other fingers. Thus hands are able to make and manipulate tools. Tools were first made at least two and a half million years ago.



Modern humans have brains approximately twice the size of those of early human toolmakers. Babies have only a quarter of the brain capacity of adults and are dependent for a long time on parental care. During this time they learn and develop complex behaviour patterns. Full development of the human brain is thought to be complete at around 18 years of age.

The shape and size of human skulls have changed over time. Early human skulls had a sloping forehead and protruding jaws. Modern skulls have a high-rounded dome, straight-planed face, and smaller jaws. The ratio of brain size to body size is larger in humans than in primates. The cerebral cortex (covering of the brain) has become more developed, as are many other parts of the brain.

Estimates vary, but around 350 000 years ago, humans used fire to their benefit, wore clothes and organised hunting of large animals. The wearing of clothes enabled humans to move into cold areas, which would have otherwise been uninhabitable for primates. It is likely that body hair was reduced in order to prevent overheating when hunting.



The Stone Age began 30 000 to 40 000 years ago and Cro-magnon humans lived as hunters and gatherers. There is evidence of art, recording events and religion from this era. This implies that there must have been speech and conventions so that social groups required for such activities could exist. Some authorities argue that from about that time the genus *Homo* began to stabilise into the one species of *Homo sapiens*.

About 10 000 years ago the New Stone Age commenced. Pottery began to be made, crops were grown, village life and trading began. Cattle were domesticated and agriculture began to replace hunting in some areas. Many people think that a major factor in the success of humans was their domestication of plants and animals. Grasses were developed into wheat, barely and rice. Cattle and sheep were used to produce milk, clothing and meat.

- 1 If you have organisms which are very similar in appearance but cannot interbreed, are they of the same species? Explain your answer.
- 2 What is *bipedalism*?
- 3 What is the shape of the human spinal column?
- 4 What does *prehensile* mean?
- 5 Which three primates most closely resemble humans?
- 6 What development allowed humans to live in cold climates?
- 7 Describe the differences between modern and early human skulls?
- 8 What is an *anthropoid*?
- 9 What advantage is there for a baby to be cared for by parents for a long time?
- 10 Why do we think that Cro-magnon humans had language and a complex social system?
- 11 Why has it been an advantage for humans to be able to walk upright?
- 12 Discuss the following statement: '*Humans are unique in that they plan, transmit skills, have complex family groups, and modify the environment.*'
- 13 What is meant by *domestication*?

Complete the following passage using the following words:

five, live, mammals, milk, primates, teeth, temperature, thumbs

Humans are 14 _____ because they have a covering of fur or hair, have 15 _____ glands to suckle their young, have a constant body 16 _____ (homeothermic) and give birth to 17 _____ young. Humans are 18 _____ because they have pentadactyl 19 (_____ -fingered) limbs, grasping hands, opposable 20 _____ well-developed binocular vision and unspecialised dentition 21 (_____).

Human evolution

Modern humans evolved from a *hominoid* (*hominid*), or human-like, species (*L. hominus* human, *oid* in the form of). From fossil evidence, scientists have come up with an order of appearance of man's ancestors. The first man-like ape was *Ramapithecus*. Remains of these were found in Miocene beds in India and Africa. They were thought to date from around 14 million years ago. The remains had reduced canines and molars and slender incisors. (Note: most sources say that it was not a hominid.) *Ramapithecus* have two main differences to apes – stance and dentition (teeth). There has been modification to the pelvis and to feet to give an upright standing position, and apes have a rectangular jaw with large canines, whereas hominids have an arched and curved jaw with reduced canines.

1 Describe the function of canine teeth.

It is generally believed that humans originated in Africa from an ancestral line (existing 6 – 4 million years ago) known as *Australopiths*. Features were a cranial (brain) size of 450 mL, small canines and incisors, very large molars and a curved jaw. Also present were large eyebrow ridges, a sloping forehead but no definite chin. It is from *Australopiths* that the hominoids later evolved. *Australopiths* were found in two forms. The gracile (more delicate and graceful) species were *Australopithecus afarensis* and *A. africanus*. A stout and more muscular group, *A. robustus* and *A. boisei*, also existed. The exact relationship between the two forms (in evolutionary terms) is still being researched. Physical anthropologists (scientists who use physical features to piece together history) continue to work toward understanding human evolution.

The earliest hominoid, an omnivore dating about 2 million years ago, was *Homo habilis*. Surviving artefacts suggest that *H. habilis* was the first human-like species to use tools and modify their environment (eg creating shelters). Not all artefacts survived, in particular those made of biodegradable materials, so it is difficult to know exactly what other tools were available. One thing is clear, the fact that rocks from distant regions are found in areas where they are uncommon, suggests that this species travelled to collect the materials they wanted for tools. Facial features were similar to *Australopithecus*. Cranial capacity was 750 mL, with molars smaller and narrower than *Australopithecus*.

2 What is cranial capacity?

3 Define artefact and give an example.

4 Define biodegradable and suggest how a biodegradable material might be made into a tool.

5 Suggest ways in which tools and modifications to their surroundings might have made life easier for *H. habilis*.

6 How might these modifications have impacted on the environment? Explain your answer.

7 Outline the function of molars.

8 Define omnivore.

Homo erectus (erect straight, raised) is a descendant of *H. habilis*. *H. erectus* lived about 1.5 million to 300 000 years ago. Of all our distant ancestors, *H. habilis*, with its larger braincase, appears to be the most similar in skeletal structure to the modern human. Overall the species was much larger than people are today. It had big bones, teeth and a much larger forehead, but the resemblance is there.

H. erectus was a hunter-gatherer and is thought to have been the first to use fire. Fire may have initially been used to repel predators and then used for cooking and land management. The numbers and arrangement of fossils found has led scientists to believe that *H. erectus* was a social species that lived and died together. *Homo erectus* (sometimes called *Sinanthropus* or *Pithecanthropus*) existed about one million years ago. Fossils have been found in Java, China, Africa and Europe. Features include a low forehead, big eyebrow ridges, smaller molars and large incisors. Brain capacity was around 1000 mL.

Much of human evolution occurred during the Pleistocene era (also called the ice age) possibly due to the drastic changes in climate that took place at the time. About 300 000 to 200 000 years ago *Homo sapiens* arrived on the scene (*sapiens* wisdom, to know). Not all groups survived and debate continues as to when speech evolved.

Homo neanderthalensis, or Neanderthal man, and Cro-Magnon man, are early human forms whose disappearances are still being questioned. Neanderthal man, named after the Neanderthal Valley Region (Rhine Province, Germany), existed about 120 000–35 000 years ago. Neanderthal man's range extended beyond Europe to Asia and Africa.

Evidence suggests that Cro-Magnon man may have been the same species as *H. sapiens*. Cro-Magnon man, dating to about 35 000 years ago, is best known for the artistic works they left on the walls of the Cro-Magnon caves (Dordogne, France). Modern humans date back to about 40 000 years ago. The type of evolution that has occurred to *H. sapiens* since this time is largely based on location and culture.

Archaic *Homo sapiens* existed in Europe (Neanderthal man), Africa and Southeast Asia around 35 000 to 120 000 years ago. With reduced molars, a slightly protruding jaw, pronounced eyebrows and a cranial capacity of 1400 mL – in most other respects was very similar to modern man. Archaic *Homo sapiens* were hunter-gatherers, had dwellings, buried the dead, used complex tools and fire.

Modern *Homo sapiens* first made an appearance about 40 000 years ago, for example Cro-Magnon man in Europe. Other fossil finds have been in North Africa, Southwest Asia and Australia. Cranial capacity was 1400 mL, a well-developed chin, high forehead and no eyebrow ridges were present. Modern *Homo sapiens* had a sophisticated culture, used complicated tools, lived in a variety of dwellings, used fire and practised burial of the dead.

- 9 Draw up a timeline with dates showing how our human ancestors appeared on Earth.
- 10 What has happened to brain size as we progress from ancient to more modern ancestors?
- 11 Why do we seem to know more about the later ancestors of humans than the earliest ones?
- 12 What are the main differences between hominids and apes?
- 13 Where did Neanderthal man exist?
- 14 Give one similarity and one difference between modern *Homo sapiens* and *Homo habilis*.
- 15 Are humans likely to evolve in the future? Explain your answer.

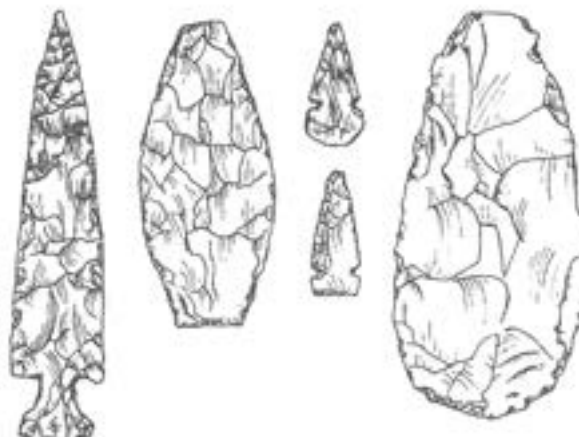
The development of human culture

All cultures have some kind of technology, social groups, arts and a spoken language.

Anthropology is the study of human beings, including their physical characteristics, culture, origins and distribution.

- 16 There is much evidence to show how human culture has developed. Study the list below and arrange in chronological order, putting the earliest example first.

- a Use of bronze.
- b Carved bone items.
- c Crops began to be grown.
- d Domestication of cattle.
- e Use of fire.
- f Industrial revolution.
- g Use of iron.
- h Paintings – often of animals.
- i Skins and wood in dwellings.
- j Stone tools.
- k Writing (on clay tablets).



The development of culture and the physical changes in hominids over time is closely related. Brain, hand and artefacts (something created by humans for a practical purpose) developed together. Language is essential to transmit technical and social skills.

Many scientists spend a lot of time comparing the lives of modern humans and early hominids. There have been many changes since humans first walked our planet, but modern humans have much in common with our ancestors.

17 Outline three points of similarity with ancient humans and three points of difference.

Main points in human evolution

Complete the following passage using these words:

agriculture, brain, chin, head, physical, posture, reduction, size, speech, tools

There are many **18** _____ changes associated with the evolution of humans. There has been an increased skull (braincase) **19** _____ and a steady increase in **20** _____ size (resulting in some reorganisation of brain structure). There has been an overall lengthening of the

21 _____ from a broad, rounded shape to a longer, oval shape and a **22** _____ in the size of the large ridges in the forehead (above the eyes). The jawbone has shortened and there has been an increase and lengthening of the **23** _____ area (previously the chin as we know it was minimal).

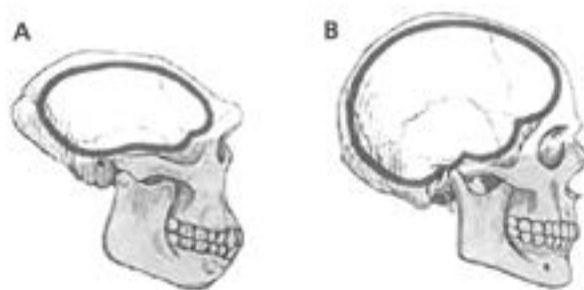
Humans have also developed an upright **24** _____ with bipedal (walking on two legs) movement.

It is not exactly known when **25** _____ was developed, but this was essential for communication.

The making of **26** _____ led to nomadic hunters and hunter-gatherers. The introduction of

27 _____ increased food availability and allowed the formation of villages and towns.

28 Correctly label the two skulls shown here as *Homo sapiens* and *Homo erectus*.



1 What is life?

- 1 Eg sea anemone, coral, plants etc.
- 2 Survival of the fittest, those that adapt or carry genetic variation survive.
- 3 Stomach, large and small intestines, gall bladder, pancreas, liver
- 4 Heart, lungs, liver, kidney
- 5 Active, recognisable shape, respond to environment, take up nutrients, reproduce, predictable lifespan.
- 6 For: dormant (sometimes for thousands of years), needs another living cell to become active. Against: once attached to living cell, actively inject their genetic material into cell, rapidly reproduce.
- 7 Deoxyribose Nucleic Acid
- 8 Fungus
- 9 (1, 2, 3) Sterile nutrient agar is placed in sterile petri dish and allowed to set. (4, 5, 6) Wire loop is sterilised by heat and used to transfer substance onto surface of agar. (7) Plate is incubated to grow any organisms.
- 10 Nutritious substance; substance which provides food.
- 11 Jelly-like material.
- 12 Surroundings/conditions that affect an organism.
- 13 Bacteria and blue green algae (cyanobacteria).
- 14 true
- 15 false
- 16 true
- 17 true
- 18 Severe form of pneumonia caused by bacteria (often in air-conditioning systems). Treated with antibiotic such as erythromycin.
- 19 Koch's postulates; discovery of cause of anthrax, TB, cholera; Nobel prize etc.

2 Prions

- 1 Hydrogen, nitrogen, carbon, oxygen and sometimes sulfur.
- 2 A state of 'sleep' where senses do not respond to any stimuli.
- 3 Inedible/unwanted parts of butchered animal.
- 4 scrapie
- 5 Bovine spongiform encephalopathy
- 6 Creutzfeldt-Jakob disease
- 7 Risk of infection by infective prions that are not identifiable.
- 8 A nerve cell.
- 9 Healthy prions.
- 10 Abnormal prion (prion protein).
- 11 Prion associated with scrapie.
- 12 To be checked.
- 13 Species with genetic material introduced into genome from another species.
- 14 To be checked.

3 Health and disease

- 1 To come in and make an action on behalf of something/someone.
- 2 Eg heart beats, breathe, alert immune system, process nutrients and remove waste.
- 3 Not sick, body functioning normally.
- 4 Something causing impaired health, illness.
- 5 Changes in body that indicate illness.
- 6 To identify a particular illness.
- 7 Eg disease, physical characteristics.
- 8 Creates genetically different daughter cells.
- 9 Red blood cells
- 10 Not fully formed and functioning.
- 11-13 To be checked.
- 14 Tiny organism, eg germ.
- 15 Destroy cells, produce toxins, interfere with normal body function, cause harmful immune response.
- 16 Mucous membrane, tears.
- 17 Protected against disease.
- 18 Disease-causing organism.

19 Toxic/poisonous substance.

20 Colourless fluid of lymph system.

4 The body's defence system

- 1 Can be penetrated.
- 2 To widen.
- 3 To separate and give off a substance.
- 4 B, A, D, C
- 5 To be checked. Include mobilise defence mechanisms vs discomfort etc.
- 6 To reduce inflammation.
- 7 Nerve cell.
- 8 No, it warns of danger, eg hot surface.
- 9 lymphocyte
- 10 cyst
- 11 macrophage
- 12 lymph
- 13 Cloning produces two or more cells containing identical DNA.
- 14 A foreign agent in the body.
- 15 Identical twins (ie formed from one fertilised egg which then separates into two.)
- 16 Stimulate B-cells and T-cells to replicate.
- 17 thymus
- 18 It means the body is more susceptible to infections and other diseases while the patient is recovering.

5 The role of white blood cells

- 1 Undifferentiated cells.
- 2 To be able to see them more clearly under a microscope; to kill them in order to keep contents still.
- 3 Granulocytes
- 4 Clear, straw-coloured liquid part of blood.
- 5 Macro large, phage eat/digest.
- 6 An insoluble blood protein.
- 7 At rest.
- 8 Monocyte: white blood cell within circulatory system. Macrophage: monocyte that left circulatory system to pursue target.
- 9 Always a 'starter' group available. Evenly spaced so closest group can work quickly as rest are activated.
- 10 Runaway immune response, risk of attacking self (eg cancer).
- 11 antigen
- 12 B: antibody production; T: terminate, helper and suppress
- 13 Antibody: immune substance, attacks antigen; Antigen: protein 'invader'.
- 14 To be checked.

6 Pathogens and antigens review quiz

- 1 dormant: at rest; immunity: protected against disease; microbes: tiny organisms, usually cause illness; antigen: foreign substance, causes illness; lymphocytes: white blood cell; antibodies: substance that attacks antigens.
- 2 antibodies, B-cells, T-cells
- 3 bone marrow
- 4 B-memory
- 5 To unite, stick together as if with glue.
- 6 Immunoglobulin, or antibodies.
- 7 They remember infections and can be quickly reactivated should the same infection recur.
- 8 thymus
- 9 It stimulates cloning in T-cells and B-cells.
- 10 T-cells
- 11 Helper T-cells
- 12 Suppressor T-cells
- 13 true 14 true 15 false 16 false 17 true 18 true
- 19 true 20 false 21 true 22 true 23 true 24 true
- 25 false 26 true

7 Robert Koch

- 1 Functional, able to work.
- 2 They could not be seen so people doubted their existence.
- 3 Destroy living organisms present.
- 4 Eg if person occupying the bed has contagious disease, to prevent reinfection and spread of disease.
- 5 Disinfecting doesn't necessarily kill all microorganisms, in particular many types of spores. Sterilising is more thorough, killing all microorganisms.
- 6 Meet nutritional needs, fluids etc. Simulate natural environment.
- 7 Research, experiments.
- 8 A type of hospital.
- 9 Most people who developed symptoms of the disease died of it.
- 10 No, only a very small fraction of people who carry the bacteria will end up with TB.
- 11 Eg exposure to greater amount of infected people than the general public.
- 12 X-rays, tuberculin test.
- 13 Bacille Calmette-Guérin
- 14 Symptoms include: cough that is worse in the morning (sometimes with blood in the sputum), chest pain, breathlessness, night sweats, and signs of pneumonia. Spread through the air when a person with untreated pulmonary TB coughs or sneezes. Prolonged exposure to a person with untreated TB usually is necessary for infection to occur.
- 15 Immune system at risk
- 16 'In vitro' literally means 'in glass', referring to organisms grown outside the body of another organism, for example in a test tube or petri dish.
- 17 Likely to contract the disease, at risk.
- 18 Illness, malnourished, premature etc. When immune system is weak.
- 19 Eg can code for (or stop) important biochemical reactions, grow out of control (cancer).
- 20 Make good basic guidelines to start experiments.
- 21 Suitable conditions for successful culturing cannot be met.
- 22 Anthrax, cholera, malaria, sleeping sickness, tuberculosis.

8 Malaria

- 1 Sweet liquid produced by plants.
- 2 Something that acts as carrier.
- 3 Organism that obtains nourishment at expense of another.
- 4 Moving, able to move freely.
- 5 Blood flows more easily.
- 6 incubation
- 7 shivering
- 8 fever
- 9 sweating
- 10 Quinine
- 11 Mosquito thrives in hot, humid conditions.
- 12 To feed her developing eggs.
- 13 Sharp, tube-like mouthpart.
- 14 Class of worm.
- 15 Mechanical: direct transfer from one organism to another. Biological: more complex, precise series of steps required.
- 16 Remove one of the steps in the required series
- 17 eg Quinine helps people who already have disease but doesn't prevent it. Insect repellent is easy to apply and effective, but can cause problems and be expensive with long term use. Netting/ mesh is relatively low tech and inexpensive, but insects can get through gaps/tears. Removing stagnant water breaks breeding cycle but difficult accomplish 100%. Insecticides can be very quick acting, but don't always kill all insects and can be environmentally damaging.
- 18 To be checked.
- 19 Sickle-cell disease is an inherited blood disorder in which the red blood cells change shape in a characteristic way. In sickle-cell disease, the haemoglobin is abnormal and this causes a change in the shape of the red blood cells.

9 SARS

- 1 Bacteria: a single-celled organism; Virus: a sub-microscopic infective agent, needs a living cell to be activated.
- 2 Viruses do not respond to antibiotics.
- 3 Rest, increase fluid intake, treat symptoms.
- 4 The strain of flu usually only affected chickens, so they were thought to be carriers.
- 5 Epidemic: national (one country); Pandemic: global (all over the world)
- 6 Eg avoid panic in community, drop in tourist revenue, reputation, exports etc.
- 7 Very bad, dangerous.
- 8 Symptoms arise quickly but are short lived.
- 9 Thought to be a mutated, different strain of flu.
- 10 Common cold, meningococcal disease etc.
- 11 Eg invaders instigate similar immune responses within the body.
- 12 Not true cells – they live off other cells.
- 13 Bacteria are true cells therefore can be destroyed, viruses are not.
- 14 To deal with other bacteria known to be present.
- 15 Eg health professionals, people working with the general public (including teachers!).
- 16 The elderly, people with respiratory complaints (asthma etc) and those with weakened immune systems.
- 17 To gain strength to recover, stop infecting other people.
- 18 Viruses causing disease have changed and vaccine and antibodies are no longer effective.
- 19 Virus: nucleic acid encased in protein coat; Viroid: virus particle, small loops of RNA, lack protein coat.
- 20 To be checked.

10 Louis Pasteur and Sir Frank Macfarlane Burnet

- 1 A hot or cold drink.
- 2 Present in absence of oxygen.
- 3 Ethanoic / acetic acid.
- 4 No, it only kills harmful microbes.
- 5 Conditions in which no microorganisms are present.
- 6 Aerobic organisms require oxygen; anaerobic organisms do not.
- 7 Anaerobic, as they would have been deprived of oxygen in the boiling water.
- 8 To kill all microorganisms present.
- 9 Fermentation: a chemical change in a substance caused by certain enzymes; Decay: to decompose or rot.
- 10 A liquid extract obtained by soaking a substance in water, eg tea.
- 11 To observe the behaviour of microorganisms in different conditions.
- 12 The cultivation of microorganisms for scientific research.
- 13 Substance prepared to inoculate against disease, usually a derivative of the disease.
- 14 Kills nearly all of a population.
- 15 Ability to resist its effects.
- 16 Microscope, so microorganisms could actually be seen.
- 17 pasteurisation
- 18 milk, wine
- 19 yeast
- 20 vinegar
- 21 anthrax
- 22 To be checked.
- 23 Gravity prevents them falling 'up' into neck.
- 24 Antibodies will store the memory of the dead virus and recognise it if it recurs.
- 25 The antibodies will be quickly activated and strengthened in fighting the live virus.
- 26 Robert Koch
- 27-28 To be checked.
- 29 Virologist: one who studies viruses; pathologist: one who studies diseases; physiology: the science of studying living organisms; immunological: dealing with the science of immunity; graftings: a portion of living tissue transplanted from



one organism to another; bacteriophages: viruses that kill bacteria; embryos: an organism in the earlier stages of development; vaccines: preventative inoculation derived from the virus; clonal: to do with clones; inherited: passed on.

30 Gain an ability not to reject foreign object.

31 Injected with virus causing disease, generates antibodies against disease that are then collected.

32 Easier, cheaper to store/house.

33 Self: part of organism, any normal cell in the body; Non-self: invader, antigen.

34 Virus attacks normal healthy cells, bacteriophage attacks bacteria.

35 The body will reject the new tissue if the immune system fights it as a foreign agent.

36 Eg rabies from dogs, malaria from mosquitos.

37 To be checked.

11 Dependence among organisms

1 Repairs to body.

2 solar

3 To do with the sea.

4 seabird

5 No, organisms usually do not only eat one thing.

6 'eaten by'

7 Seaweed, small shellfish, lobster, seal.

8 Lobster, crab, prawn.

9 Autotroph produces its own food; Heterotroph relies on other organisms for food.

10 Leaves/vegetation → insects → small birds → raptors

11 leaves/vegetation

12 raptor/eagle/hawk

13 bird of prey

14 Phytoplankton → insect larvae → fingerlings → trout

15 phytoplankton

16 insect larvae

17 fingerlings

18 trout

19 Fingerlings would increase, as they are not being eaten by trout.

20 Tiny aquatic plants.

21 'Biological mass' – matter that is living or has been living.

22 To build up stocks for fishing.

12 Flows of energy through ecosystems

1 Feeding relationships between organisms in an ecosystem.

2 grass, tree

3 sea grasses, seaweed

4 eg wallaby, kangaroo

5 Carnivore if hunting fish but a secondary (top) carnivore if hunting the fish-eating (carnivorous) birds. A tertiary carnivore would eat the skua, or the skua itself could be considered a carnivore.

6 Meat: carnivore; plant: herbivore.

7 omnivore

8 To be checked.

13 Components of an environment – biotic and abiotic

1 Non-living components, eg rocks, water, air.

2 The salt concentration in the water.

3 Most water is locked up in the form of ice, very little vegetation.

4 Very dry.

5 Thick overhead tree cover.

6 grass eater

7 Trees that grow with roots above ground because they are in coastal swamps.

8 Ridge of coral or rock just below sea's surface.

9 Dense tropical forest in heavy rain area.

10 Spinifex: grass found in Australia, East Asia;

Tussock: dense tufts of grass.

11 Sclerophyll 'bush' has continual vegetation growth, no dormant period. Dry – limited rain; wet – more and more regular rain.

14 Bioaccumulation and biomagnification

1 1×1 per bug = 1, 1 mini x 50 bugs = 50,

1 killer x 10 mini = 500, 1 bird x 5 killer = 250,

1 carnivore x 3 bird = 7500

2 Taking something from the environment into the organism.

3 Take in for digestion.

4 Putting away for future use.

5 To get rid of.

6 State of being in balance.

7 Bioaccumulation: how substances enter food chain;

Biomagnification: how the effects of accumulation are magnified as they move through food chain.

8 Eg levels in sediment, plant tissue, water.

9 Study of cycling of toxins through an ecosystem.

10 trophic level

11 carnivore

12 nutrient

13 ecosystem

15 Heavy metal pollution: the Minimata case

1 Cheaper than safe disposal costs and possible future compensation claims.

2 Less biomass so more concentrated.

3 Probably not, because same fish would not be repeatedly exposed to toxin which would also wash out to sea. As inland body, fish are contained in toxic water.

4 Employment prospects besides fishing, prosperity.

5 Thermometers, barometers, electrical switches, lighting, batteries and in the manufacture of sodium hydroxide.

6 Top carnivores so would contain highest concentrations of toxin.

7 Mercury used to treat fur/felt in hats caused mental illness.

16 DDT: the Peregrine Falcon case

1 Chemical used to kill insects.

2 Time required for half of the molecules to break down.

3 It ate many toxic fish or insects.

4 Kills more than just target species, not selective.

5 Accumulates in fatty tissues.

6-7 To be checked.

17 Parasitism, mutualism and commensalism

1 Mutualism, as both organisms benefit.

2 mutualism

3 Have cellulose broken down for them.

4 Feed on the cellulose.

5 Parasites – they benefit, but the host suffers.

6 parasite (often called 'social parasite')

7 host

8 ruminants

18 Behavioural adaptations to environment

1 Those organisms that are more flexible are likely to survive changed conditions and pass on beneficial adaptations to offspring.

2 Adapted: adjusted to conditions; Inherited: acquired from ancestry.

3 Eg a change in feeding behaviour to maximise a new food. Those most willing to change behaviour (eg scratch for food vs actively dig) most likely to survive.

4 Individuals best suited to exploit it will generally have higher survival rate so produce more offspring. This increases trait in each new generation.

5 They ensure that the strongest members of their species survives to reproduce, rather than all dying due to lack of food.

6 Eg don't need to be powerful if you work in numbers (ants, bees etc).

7 Eg single parent vs two parents vs communal care of young.

8 cannibal

9 habitat

10 evolution

11 Inheritance is determined at cellular level, and cannot be altered by physical changes.

19 Fossils reviewed

1 Make stiff, rigid.

2 elephant

3 Both deal with relationships between fossils, land/rock formations.

4 Ice freezes tissue, microorganisms that would break it down cannot survive in the cold conditions.

5 No oxygen, smothered.

6 Hard pieces, bark, seed.

7 Relative sizes based on comparing with other fossils, depth of footprints and their distance apart indicate size and weight of dinosaur.

8 Hollow mould formed by imprint. Shows where original material/fossil was.

9 Uses rock layers: age of layer indicates fossil age.

10 Original material forming the fossil is gradually replaced with other until new material takes its place in the form.

11 Imprints, eg footprint in sand.

20 Strata and fossils

1-4 To be checked.

5 8

6 8

7 Lagganopod: at the bottom layer.

8 Bindapid: at the top.

9 Layers shifted, geological activity.

21 Vertebrate evolution

1 Kingdom, Phylum, Class, Order, Genus, Species

2 Evolve at the same time.

3 D

4 Two equal sides, mirror images.

5 Symmetry around a central point or axis.

6 Urochordata.

7 Soft bodies so less chance of fossilising.

8 To be checked. Lamarck believed acquired characteristics could be inherited etc.

22 Primates

1 insect eater

2 apes (gorillas), baboons

3 Better depth and distance perception.

4 tree dweller

5 climb, manipulate

6 Walking on two legs.

7 To be checked. Eg opposable thumb, insectivore (some omnivore), 30-40 cm. Student to use logical argument.

8 grasp, balance

9 It is safer, hidden, away from predators.

10 Hands that can operate and control objects.

11 Taking stances with behavioural significance.

12 This is generally the case, as the offspring take longer to learn how to be independent.

13 Koko was taught to communicate with humans using sign language.

14 The sloth is not a primate.

23 Primates revision questions

1 Apes and monkeys.

2 Walk on all fours.

3 Americas

4 Eg flatter face, two premolar teeth, areas on body/face without hair.

5 Great apes

6 vision

7 gorilla

8 All great apes (chimpanzee, orang-utan, gorilla).

9 About 3 million years ago

10 Posture, sound, facial expression, signals.

11 true 12 false 13 true 14 true 15 true 16 true

17 true 18 false 19 true 20 true 21 true 22 true

23 false 24 false 25 true 26 true 27 false 28 false

29 true 30 false

24 Homo sapiens

1 No, chromosomes would be too different to pair up correctly.

2 Walk on two legs.

3 S-shaped

4 grasping

5 Chimpanzee, gorilla, orang-utan.

6 Wearing clothes, use of fire.

7 Doubled brain size, no more sloping forehead or protruding jaw, modern has high, rounded dome shape.

8 Suborder of primates that resemble humans.

9 Increases survival rate.

10 Left evidence of art and religious events.

11 Can see further, frees hands to manipulate objects.

12 To be checked, should give examples where other animals do at least some of these.

13 Changing (tamed) so that they can be used by humans more easily.

14 mammals

15 milk

16 temperature

17 live

18 primates

19 five

20 thumbs

21 teeth

25 Human evolution

1 To tear (usually meat).

2 Skull volume.

3 Objects made by humans, eg clay pots, jugs, jewellery.

4 Able to break down in environment, eg wooden axe handle.

5 Eg kill/cut food, make shelter and clothing.

6 To be checked, should relate to objects chosen.

7 Grinding teeth.

8 Eats meat and vegetation.

9 To be checked.

10 increased

11 More surviving artefacts/bones etc.

12 Hominids resemble modern humans more closely.

13 In Europe, around what is now Germany.

14 Similarities: eg used tools and modified environment. Differences: brain size, use of agriculture etc.

15 Probably not. More likely to change environment than to adapt to changes.

16 (Note: There may be variations in order given) j, b, e, i, h, c, d, k, a, g, f

17 Eg need to have food, shelter, safety etc, technology, dependence on outside sources for food, clothing.

18 physical

19 size

20 brain

21 head

22 reduction

23 chin

24 posture

25 speech

26 tools

27 agriculture

28 A *Homo erectus* B *Homo sapiens*

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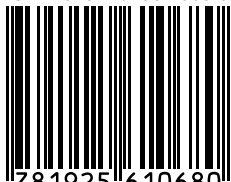
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